

Where we are...

Assignment 1 is due this week; you can use AI in whatever way you want

Please finish your **Labs 4 & 5** this week

Lab 6 (next Wed) is an in-class activity; after that 🌸

An **extra credit assignment** will be released this week; due May 6 2026

Animation

CMSC471: Introduction to Information Visualization

Fumeng Yang, Assistant Professor

March 2, 2026

University of Maryland, College Park

<https://fumeng-yang.github.io/CMSC471>

Animation

Likes

Understand a state transition

Direct attention

Visual variable to encode data

Explain a process

Increase engagement/entertainment

Interaction



Storytelling

Struggles

Cognitive overload

Distracting/confusing

Perceived as unscientific

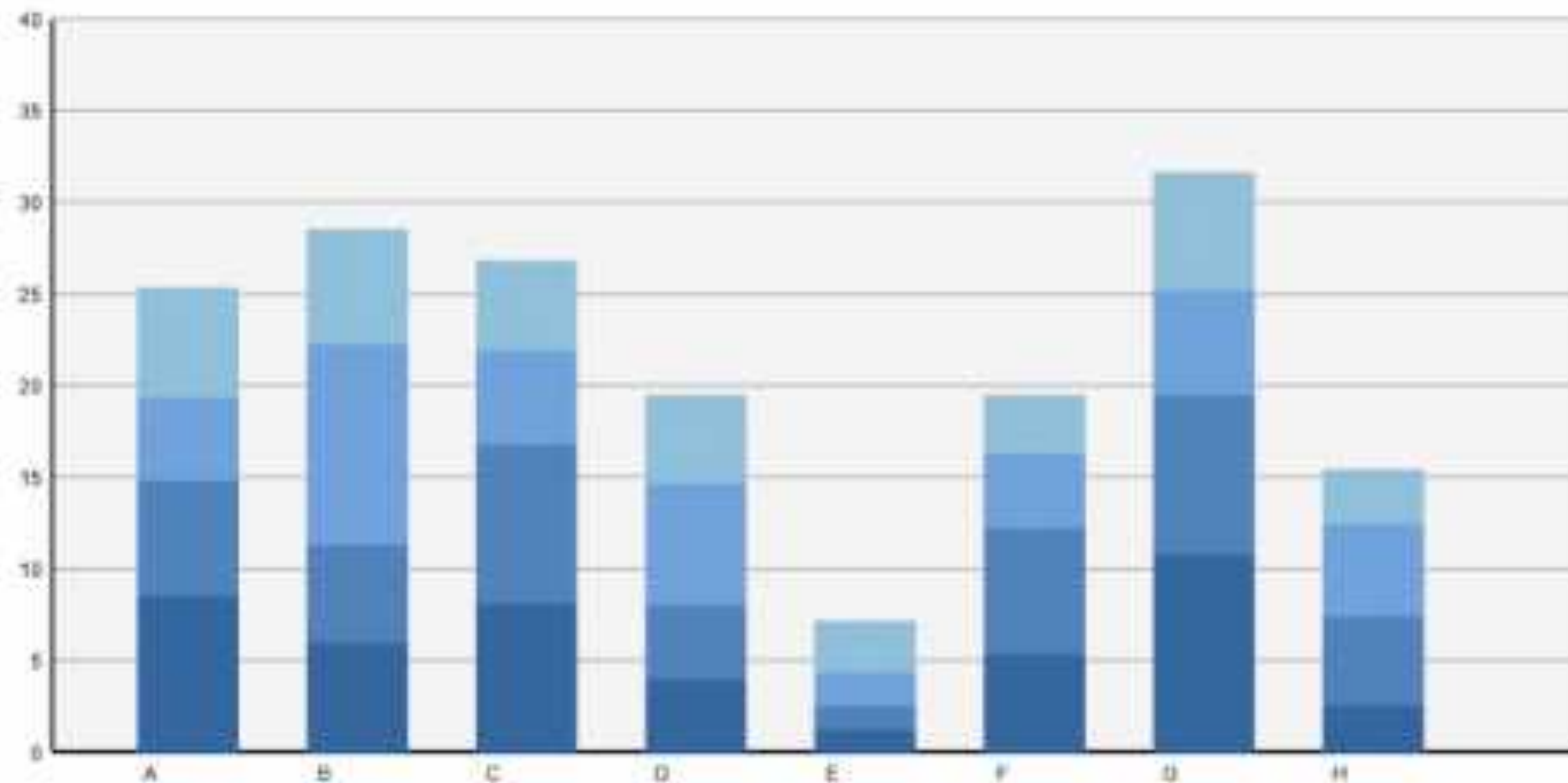
Tedious

Implementation/performance

Animations we like

Understand a state transition

All can be
implemented in
D3.js

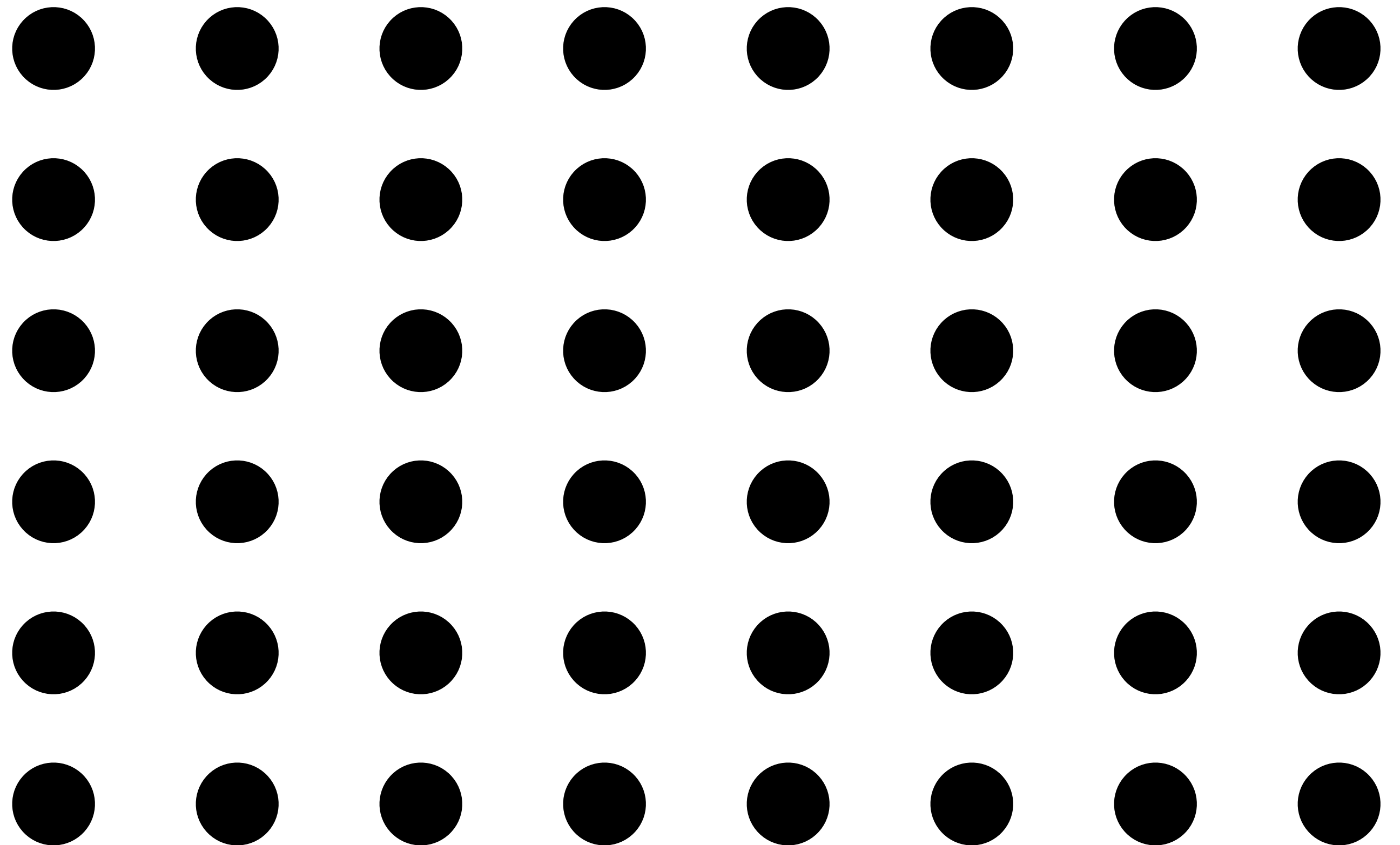


Direct attention

Motion as a visual cue

Pre-attentive, stronger than
color, shape, etc.

I use it a lot in my slides
to guide your attention



Variable

wind map

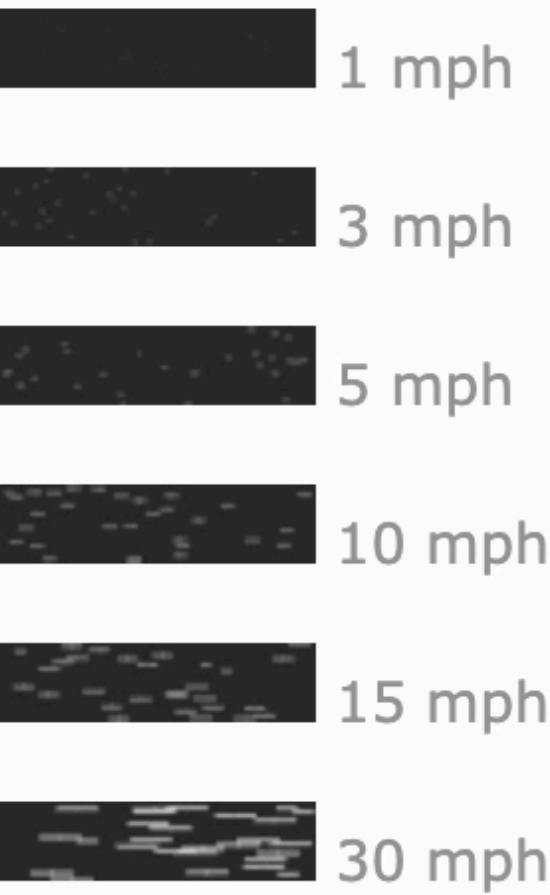
Nov. 8, 2024

7:12 pm EST

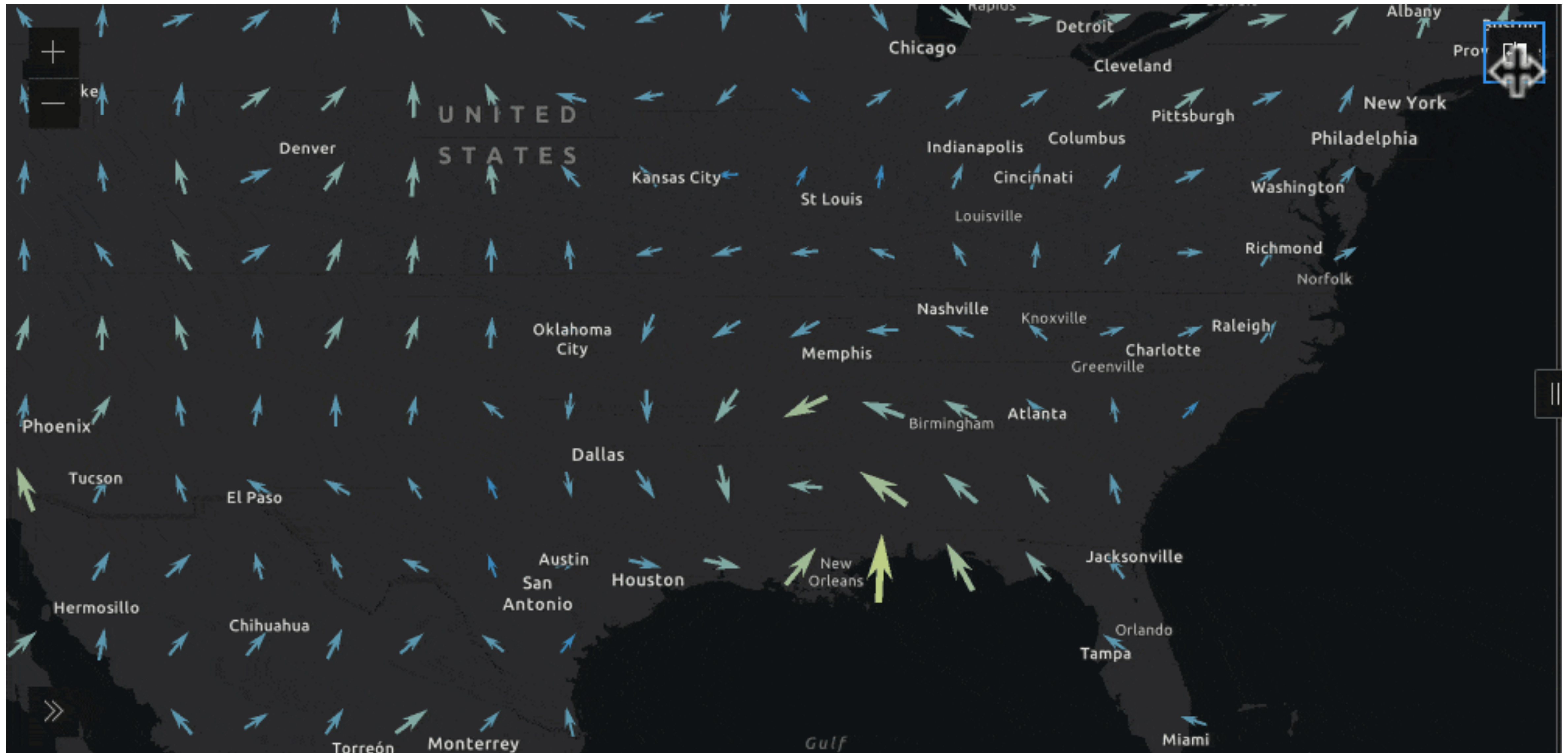
(time of forecast download)

top speed: 62.2 mph

average: 7.2 mph



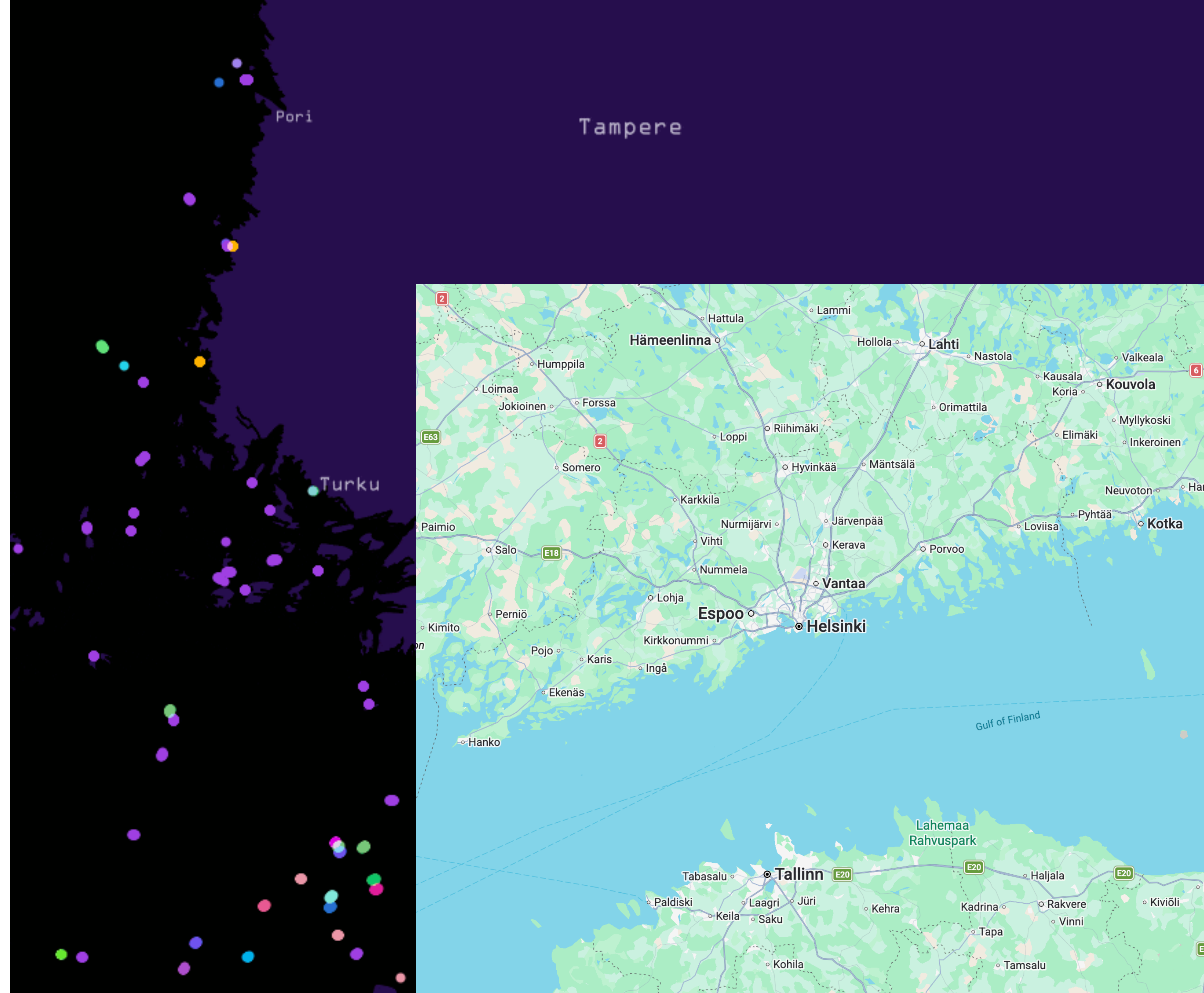
... probably better than this



Variable

24 hours of marine traffic on the Gulf of Finland. Highlights multiple interesting facts e.g. how important trade route the Baltic Sea is, how many ferries there are in the Turku archipelago and how Finns really like visiting Estonia. Each dot represents one ship. Colors represent the countries of origin.

<https://tjukanov.org/gulfoffinland/>

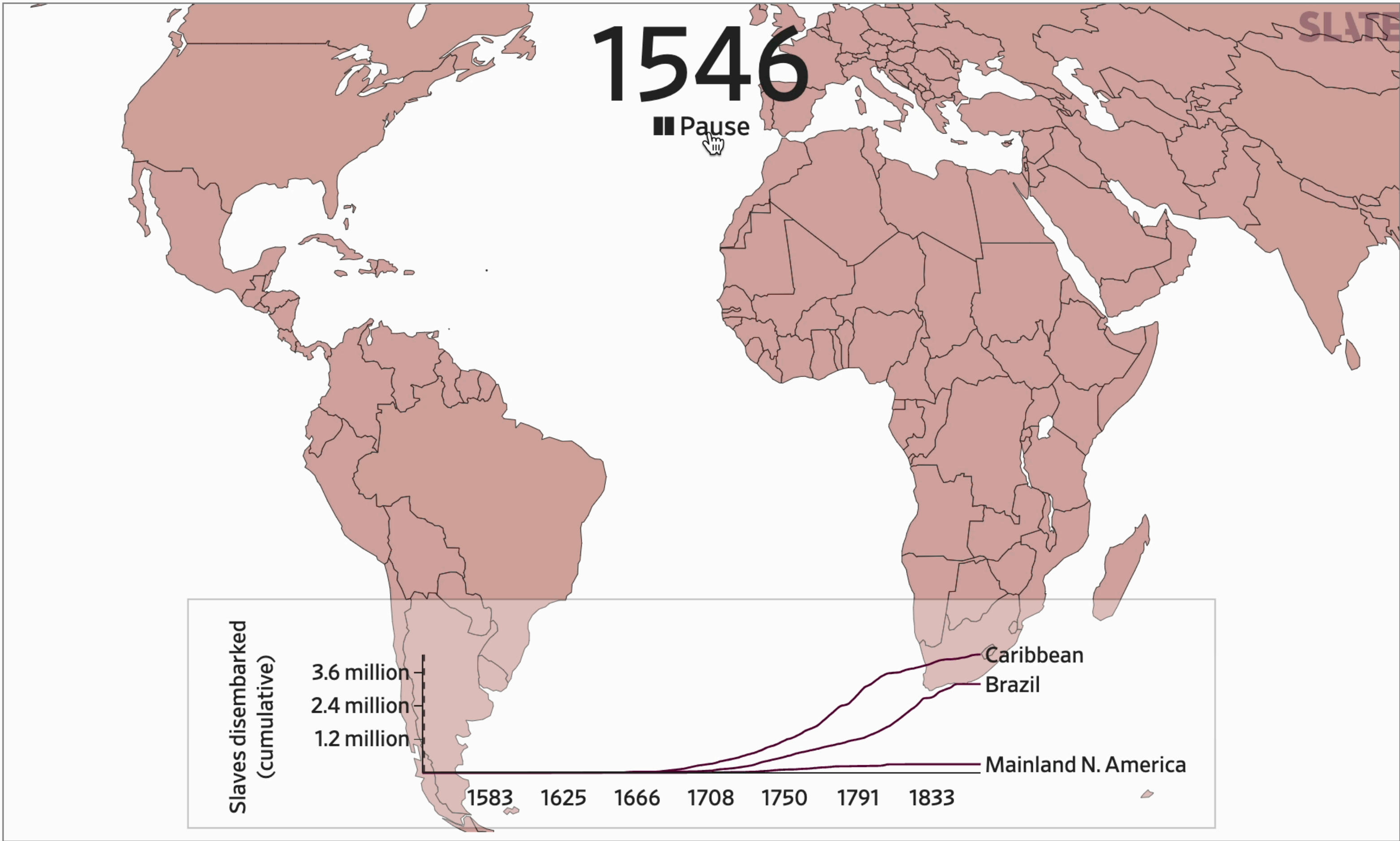


Variable

The Atlantic Slave Trade in Two Minutes

315 years. 20,528 voyages. Millions of lives.

BY ANDREW KAHN AND JAMELLE BOUIE SEPT 16, 2021 • 4:18 PM



<https://slate.com/news-and-politics/2021/09/atlantic-slave-trade-history-animated-interactive.html>

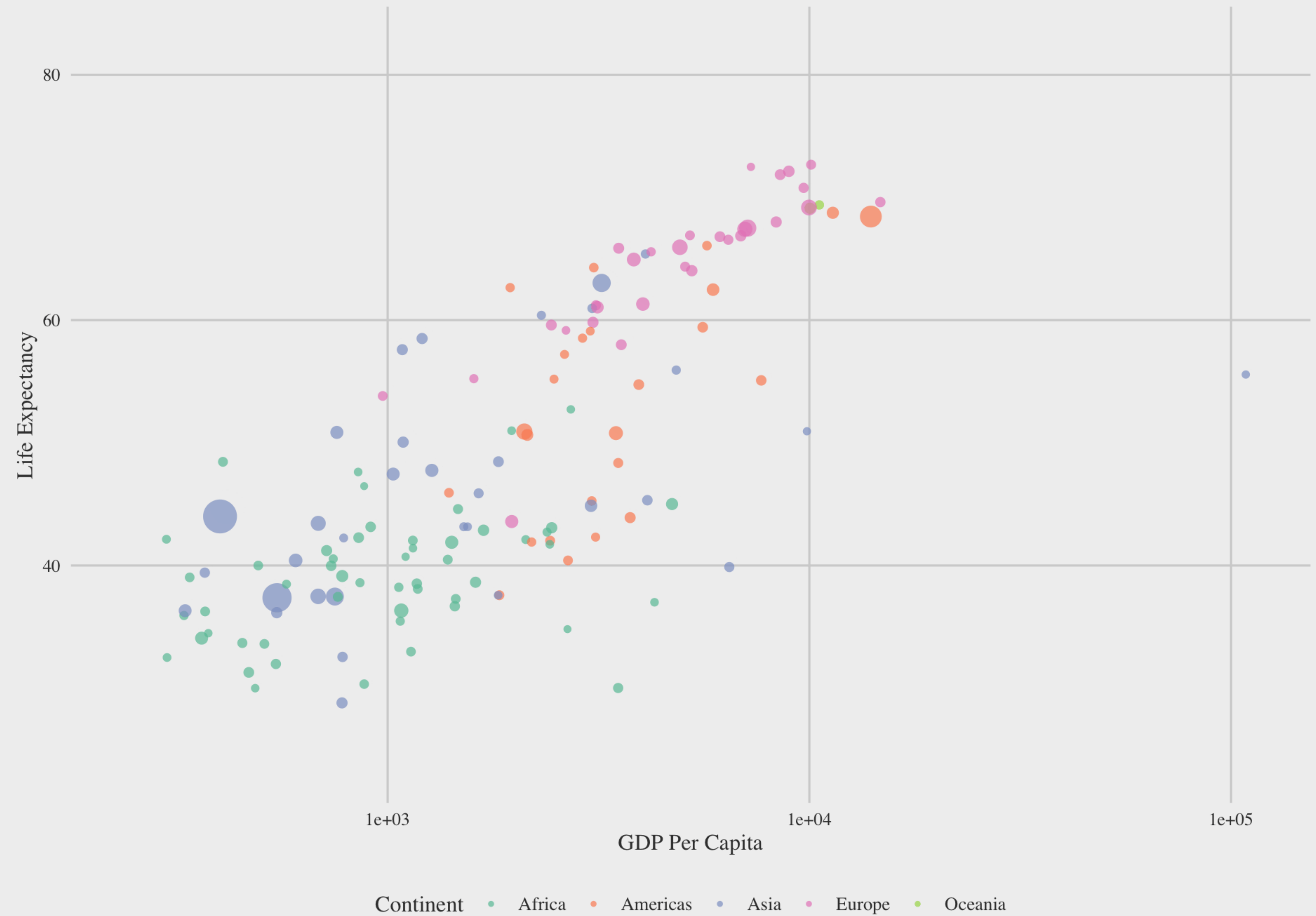
Variable



vs. this

Life Expectancy vs GDP per Capita by Country

Year: 1952



Variable

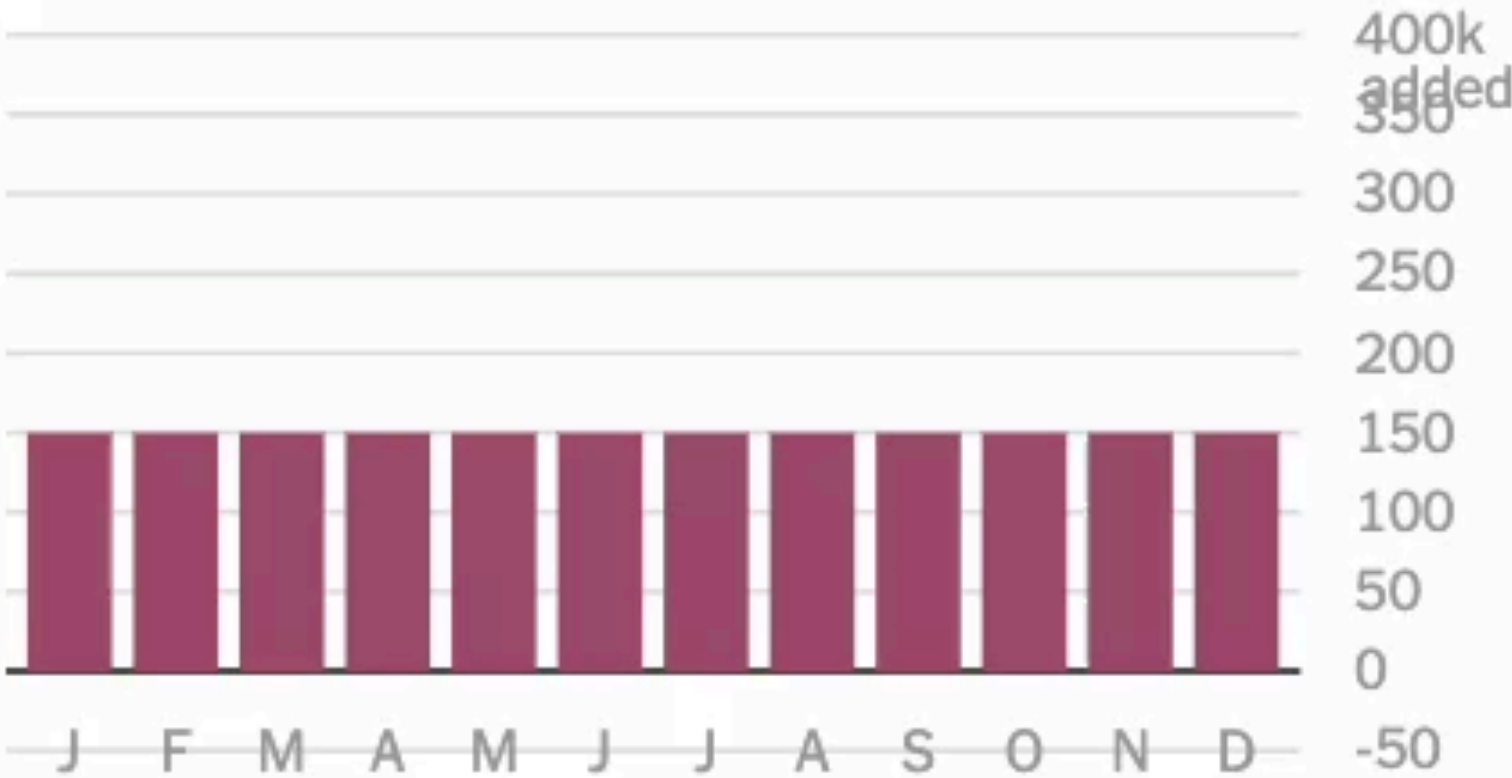
Show uncertainty by
looping over likely
outcomes

<https://www.nytimes.com/2014/05/02/upshot/how-not-to-be-misled-by-the-jobs-report.html>

How Not to Be Misled by the Jobs Report

Let's consider how statistical noise might affect our perception of the jobs report in two scenarios: one with steady job growth and one with accelerating job growth.

If job growth **were actually steady**
over the last 12 months...



...the jobs report
could look like this:

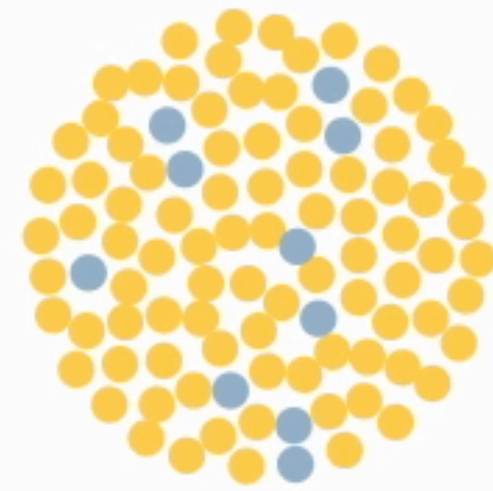


Did you infer a pattern from the chart on the right? If you did, you were reading too much into the jobs numbers. That chart should be a straight line, but we've added the same amount of sampling error that the jobs report has.

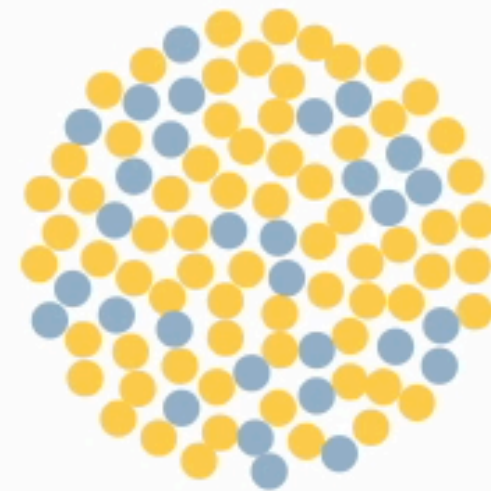
Variable / Process

Watch how the measles outbreak spreads when kids get vaccinated – and when they don't

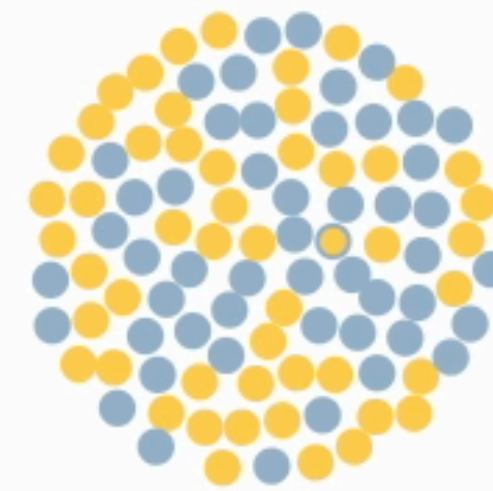
😊 vaccinated 😊 susceptible 😊 vaccinated but susceptible 😊 infected ● contact with an infected person



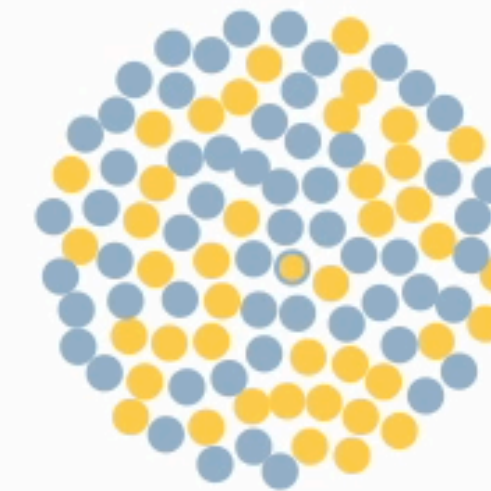
10.0% vax rate



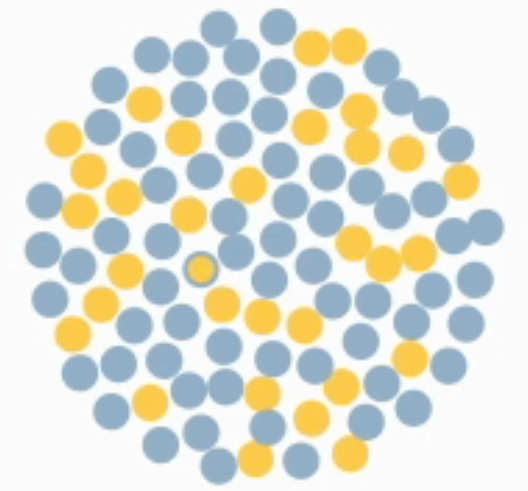
30.0% vax rate



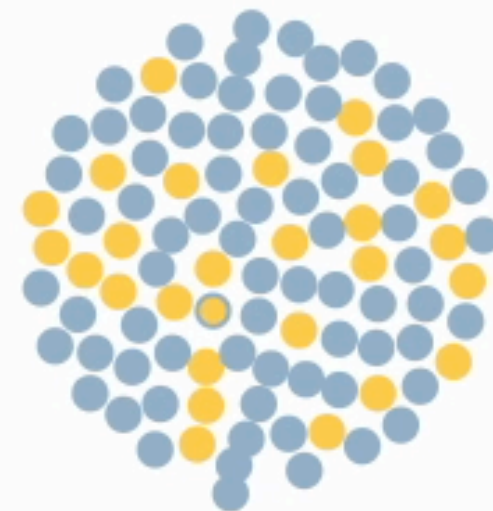
50.0% vax rate



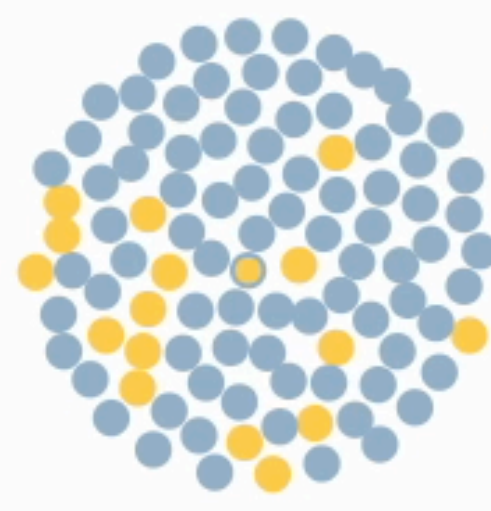
58.5% vax rate, similar to
Okanagan County, WA



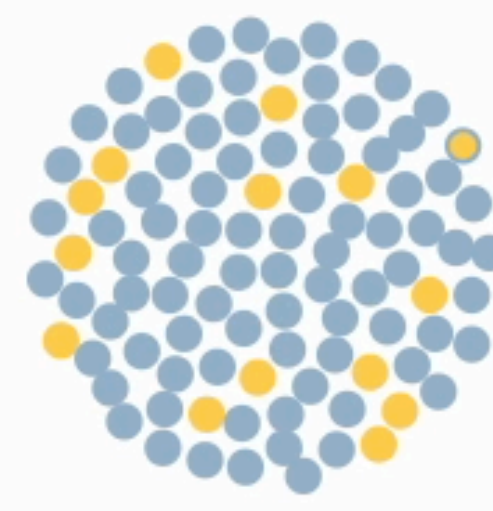
68.9% vax rate, similar to
Thurston County, WA



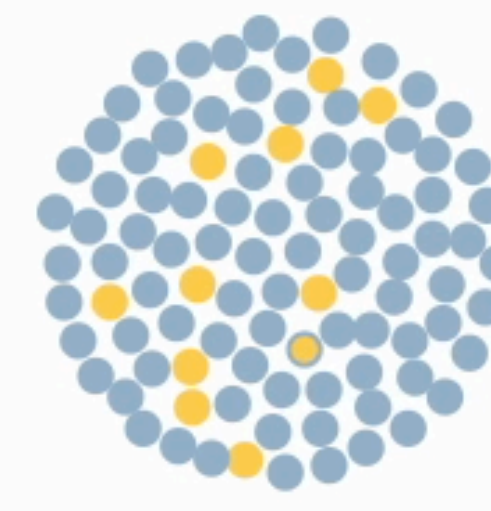
74.4% vax rate, similar to
Island County, WA



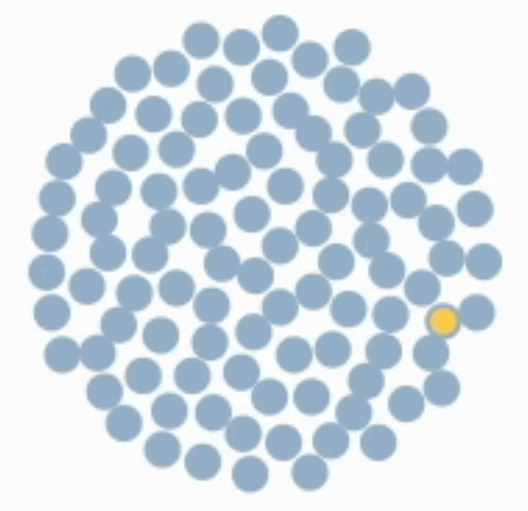
83.8% vax rate, similar to
Santa Cruz County, CA



86.0% vax rate, similar to
Los Angeles County, CA



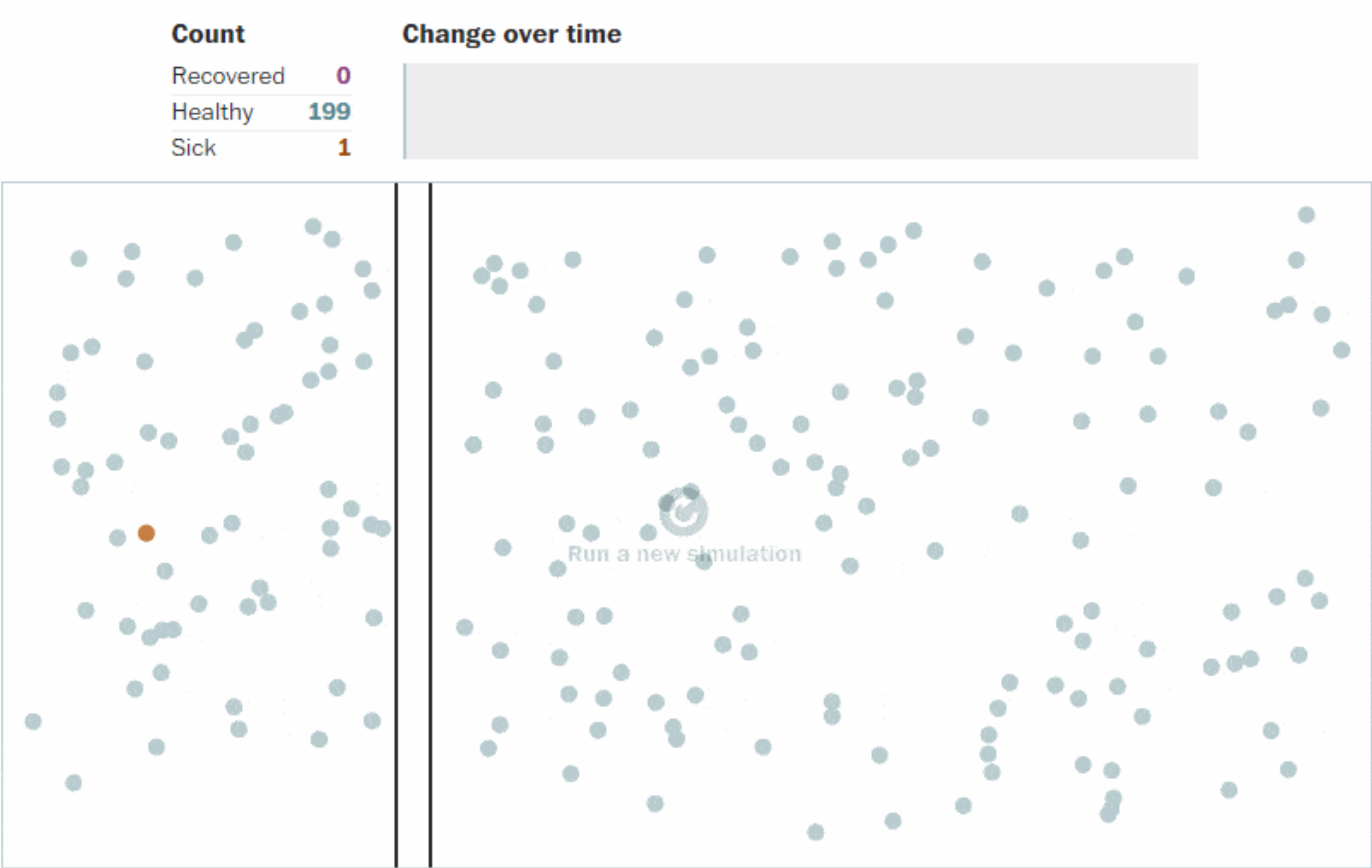
90.0% vax rate, similar to
Orange County, CA



99.7% vax rate, similar to
Gadsden County, FL

<https://www.theguardian.com/society/ng-interactive/2015/feb/05/-sp-watch-how-measles-outbreak-spreads-when-kids-get-vaccinated>

Variable / Process



Explain a process

The Bootstrap

Much of frequentist inference centers on the use of "good" estimators. The precise distributions of these estimators, however, can often be difficult to derive analytically. The computational technique known as the Bootstrap provides a convenient way to estimate properties of an estimator via resampling. In this example, we resample with replacement from the empirical distribution function (which is itself generated by sampling once from the population) in order to estimate the standard error of the sample mean.

Choose a probability distribution from which we will sample once to generate the empirical distribution function.

-- select a distribution --

Choose a sample (and resampling) size (n) and sample from your chosen distribution.

n = 10

Sample

Resample to get an idea of the spread of the sample mean's distribution.

Resample

Resample 100 times

distribution

sample

resample + average

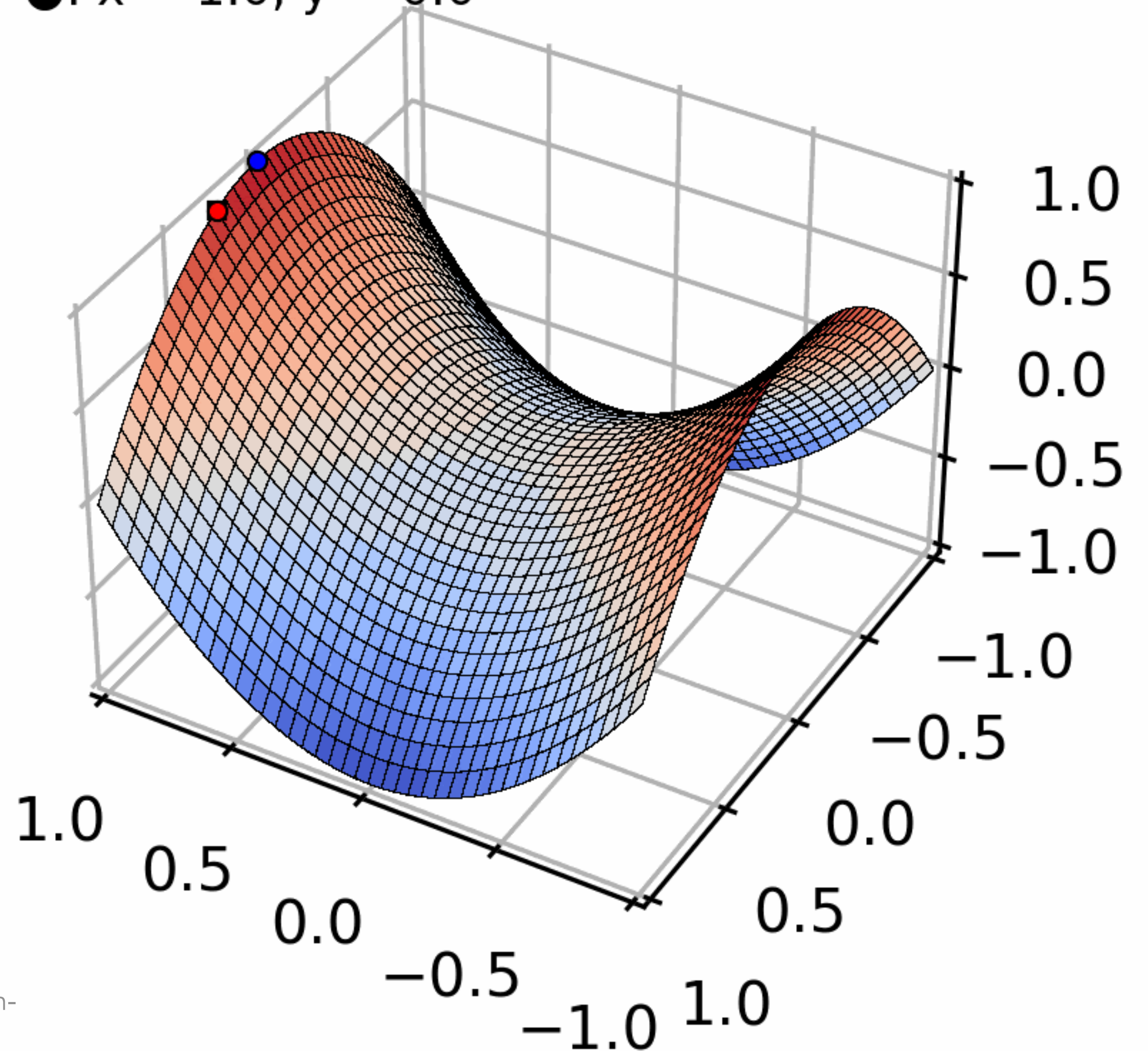
count

Explain a process

Initialization

■: $x = 1.0, y = 0.25$

●: $x = 1.0, y = 0.0$



Explain a process

<https://www.netlogoweb.org/launch#http://www.netlogoweb.org/assets/modelslib/Sample%20Models/Biology/Wolf%20Sheep%20Predation.nlogo>

Search the Models Library:

Sample Models/Biology/Wolf Sheep Predation

Upload a Model:

Choose File

No file chosen

powered by NetLogo

Wolf Sheep Predation

File: [New](#) [Revert to Original](#)

Export: [NetLogo](#) [HTML](#)



Mode: Interactive

Commands and Code: Bottom

model speed

ticks:

model-version
sheep-wolves

initial-number-sh... 100 initial-number-wo... 50

grass-regrowth-time 30

setup

go

Sheep settings

sheep-gain-from-food 4

sheep-reproduce 4 %

Wolf settings

wolf-gain-from-f... 20

wolf-reproduce 5 %

☐ show-energy?

sheep

0

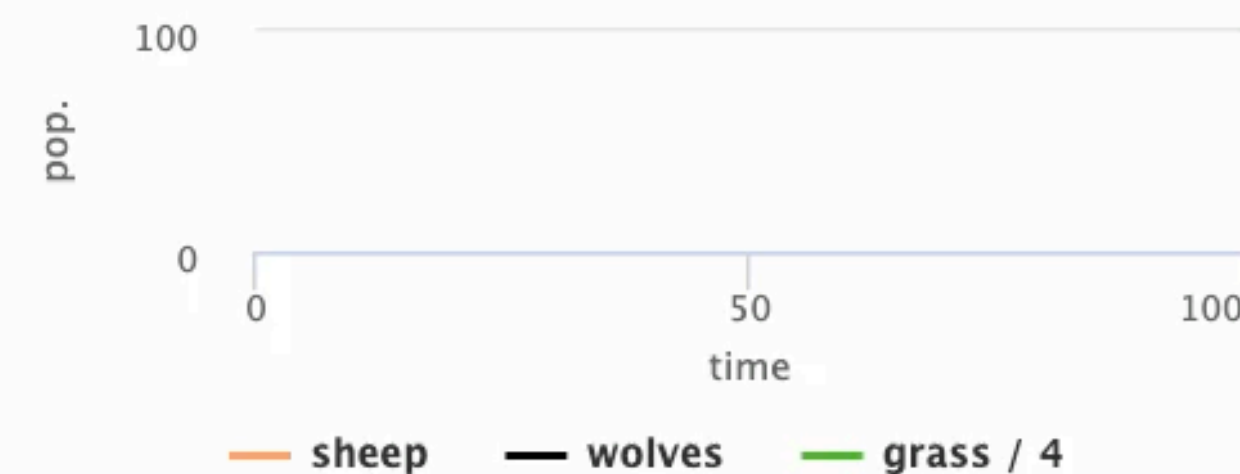
wolves

0

grass

N/A

populations



Apple's Value Hit \$1 Trillion. Add Disney to Bank of America and ... You're Halfway There

By JON HUANG, KARL RUSSELL and JACK NICAS AUG. 2, 2018

START

Engagement

Extensive Data Shows Punishing Reach of Racism for Black Boys

By EMILY BADGER, CLAIRE CAIN MILLER, ADAM PEARCE and KEVIN QUEALY MARCH 19, 2018

Black boys raised in America, even in the wealthiest families and living in some of the most well-to-do neighborhoods, still earn less in adulthood than white boys with similar backgrounds, according to a sweeping new study that traced the lives of millions of children.

White boys who grow up rich are likely to remain that way. Black boys raised at the top, however, are more likely to become poor than to stay wealthy in their own adult households.

Follow the lives of 130 boys who grew up in rich families ...



Grew up rich

Most white boys ■ raised in wealthy families will stay rich or upper middle class as adults, but black boys ■ raised in similarly rich households will not.

...and see where they end up as adults:

WHITE MEN BLACK MEN

Rich adult

0

0%

0

0%

Upper-middle-class adult

0

0%

0

0%

Middle-class adult

0

0%

0

0%

Lower-middle-class adult

0

0%

0

0%

Poor adult

0

0%

0

0%

<https://www.nytimes.com/interactive/2018/03/19/upshot/race-class-white-and-black-men.html>

Income Mobility Charts for Girls, Asian-Americans and Other Groups. Or Make Your Own.

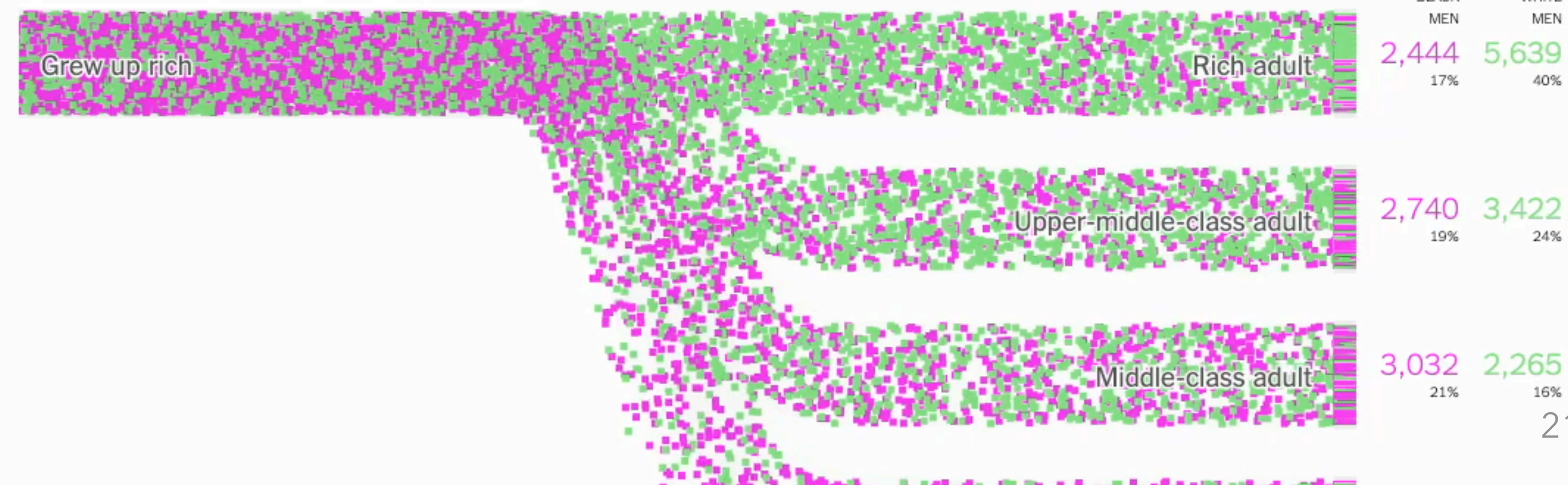
By EMILY BADGER, CLAIRE CAIN MILLER, ADAM PEARCE and KEVIN QUEALY MARCH 27, 2018

Last week we [wrote](#) about a sweeping new study of income inequality, which followed 20 million children in the United States and showed how their adult incomes varied by race and gender. The research was based on data about virtually all Americans now in their late 30s.

This first animated chart illustrates one of the study's main findings: White boys who grow up rich are likely to remain that way, while black boys raised in similarly wealthy households are more likely to fall to the bottom than stay at the top in their own adult households.

Black and **white** boys raised in wealthy families

Follow the lives of these **41,877** Americans and see where they end up as adults:



Group activity

Scroll all way down

<https://www.nytimes.com/interactive/2018/03/27/upshot/make-your-own-mobility-animation.html>

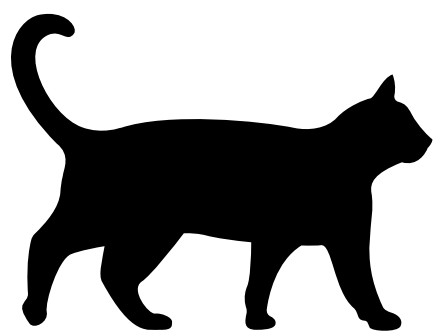
Select a combination, and share your thoughts

**Animations we
struggle with**

Distracting

Animations can be distracting

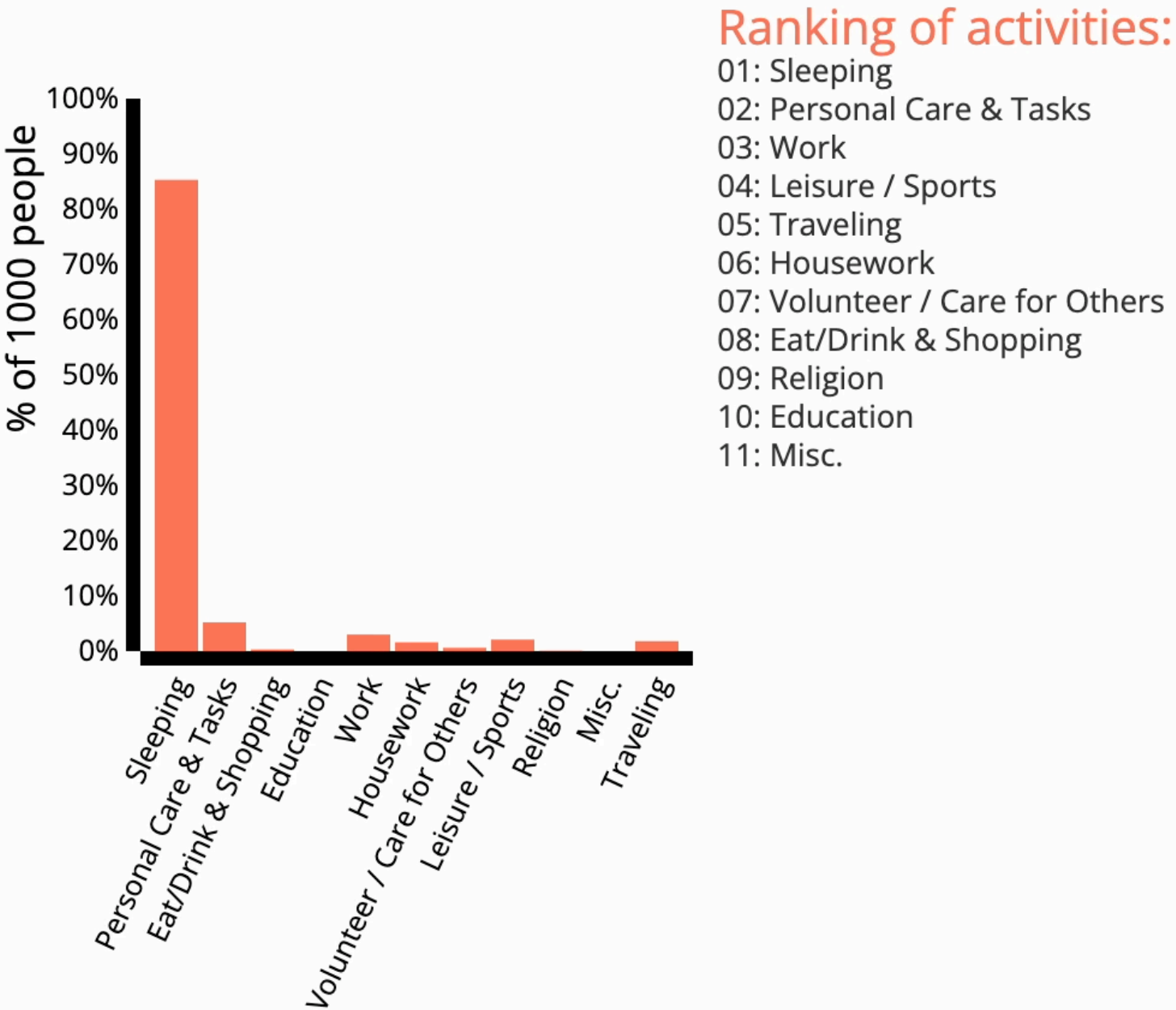
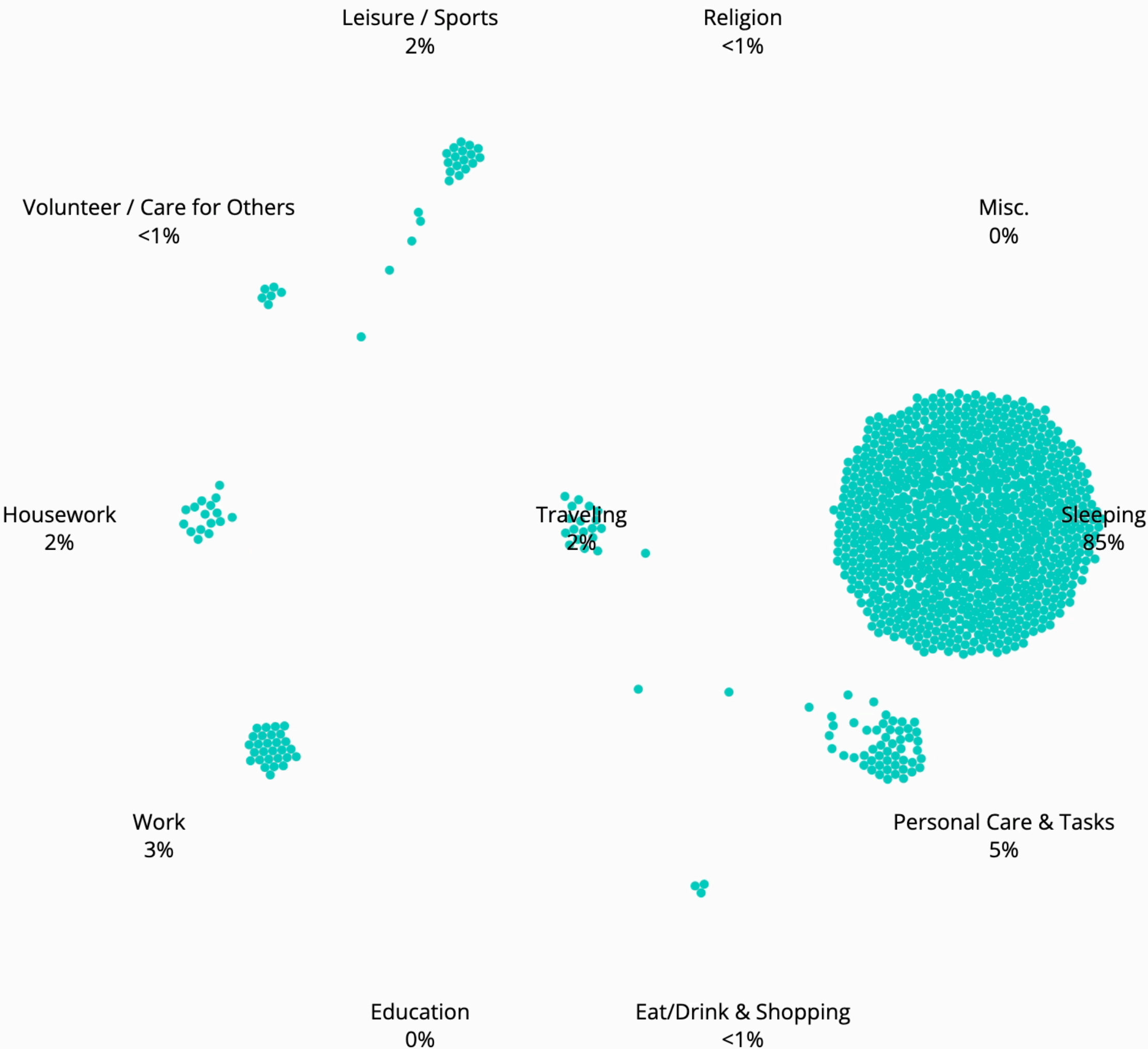
Were you listening to me?



Cognitive overload

4:12am

⏮ ⏪ SLOW FAST ⏩ ⏭



Not that I don't like this visualization, it's really a lot

Tedious

4:27am

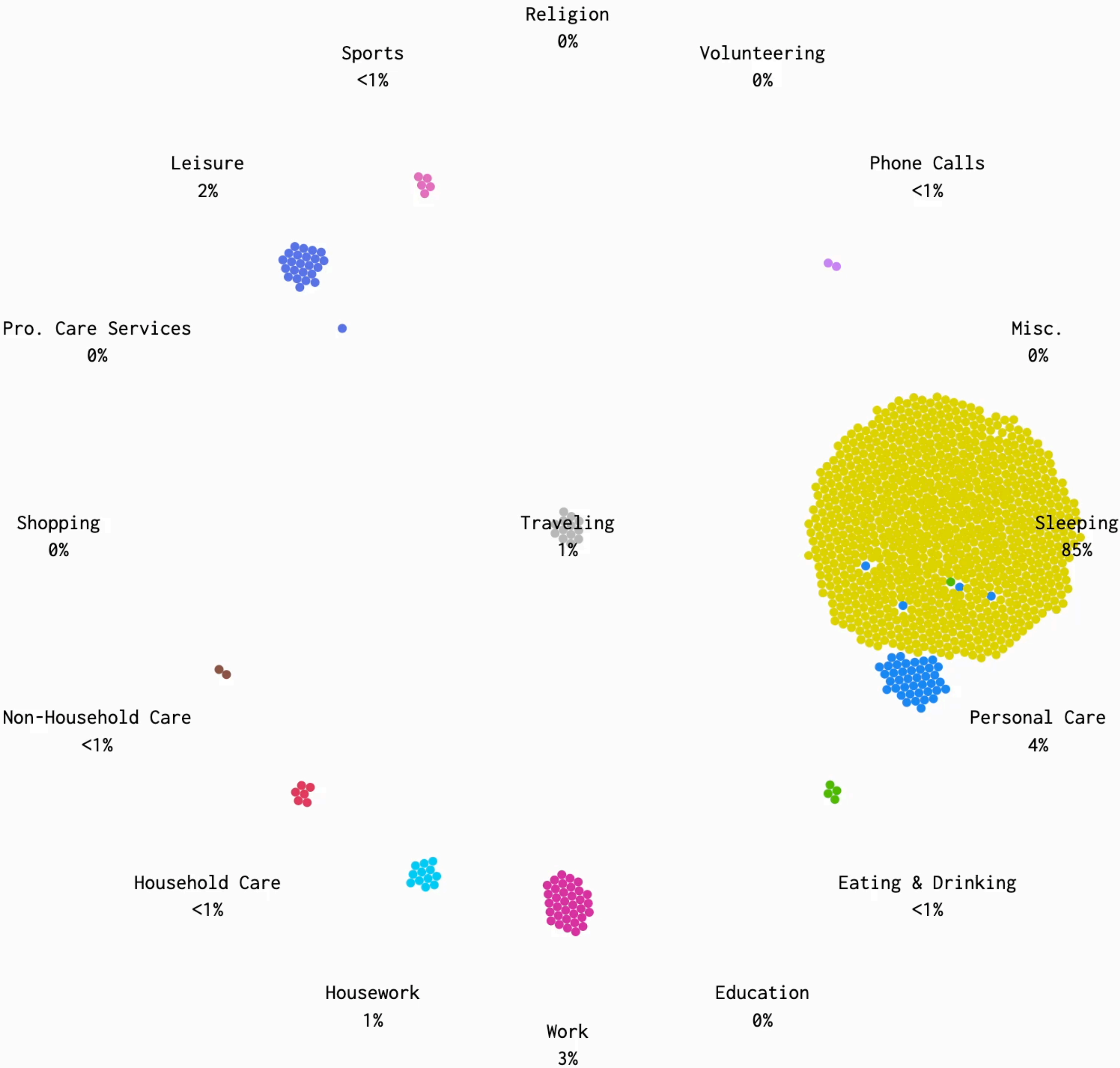
SLOW

MEDIUM

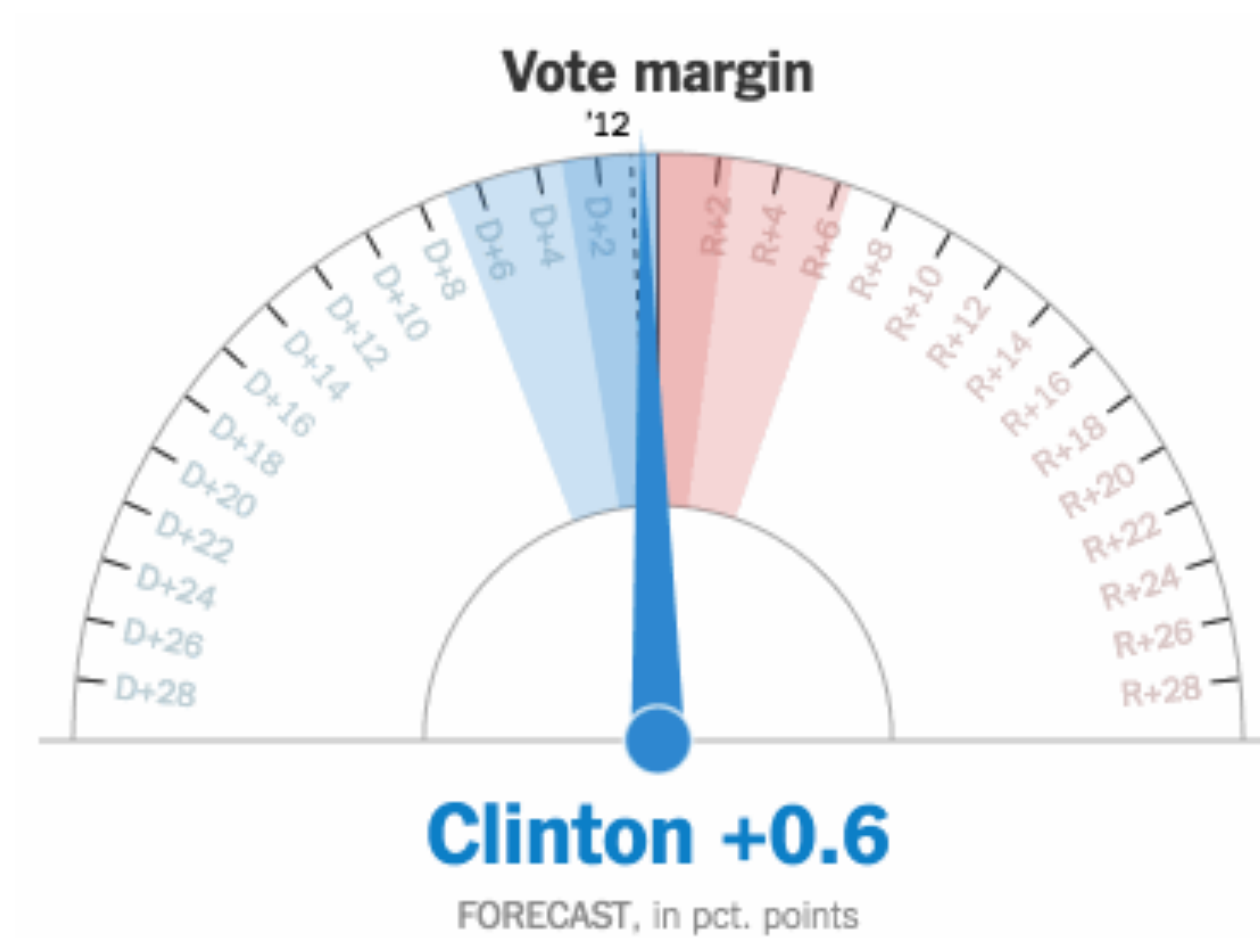
FAST

The simulation kicks in, based on data from the American Time Use Survey.

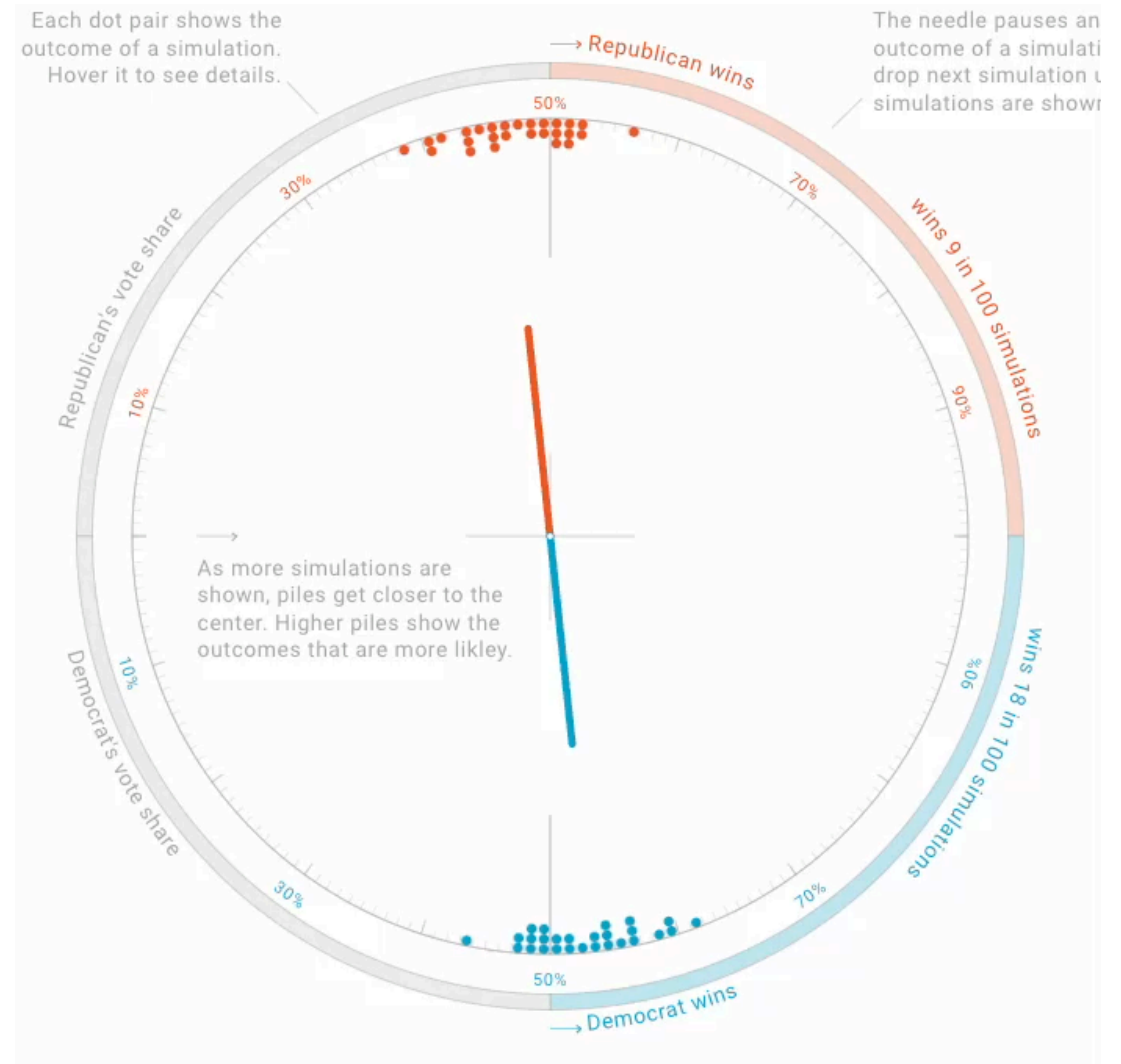
This is a simulation of 1,000 people's average day. It's based on 2014 data from the [American Time Use Survey](#), made way more accessible by the [ATUS Extract Builder](#).



Confusing



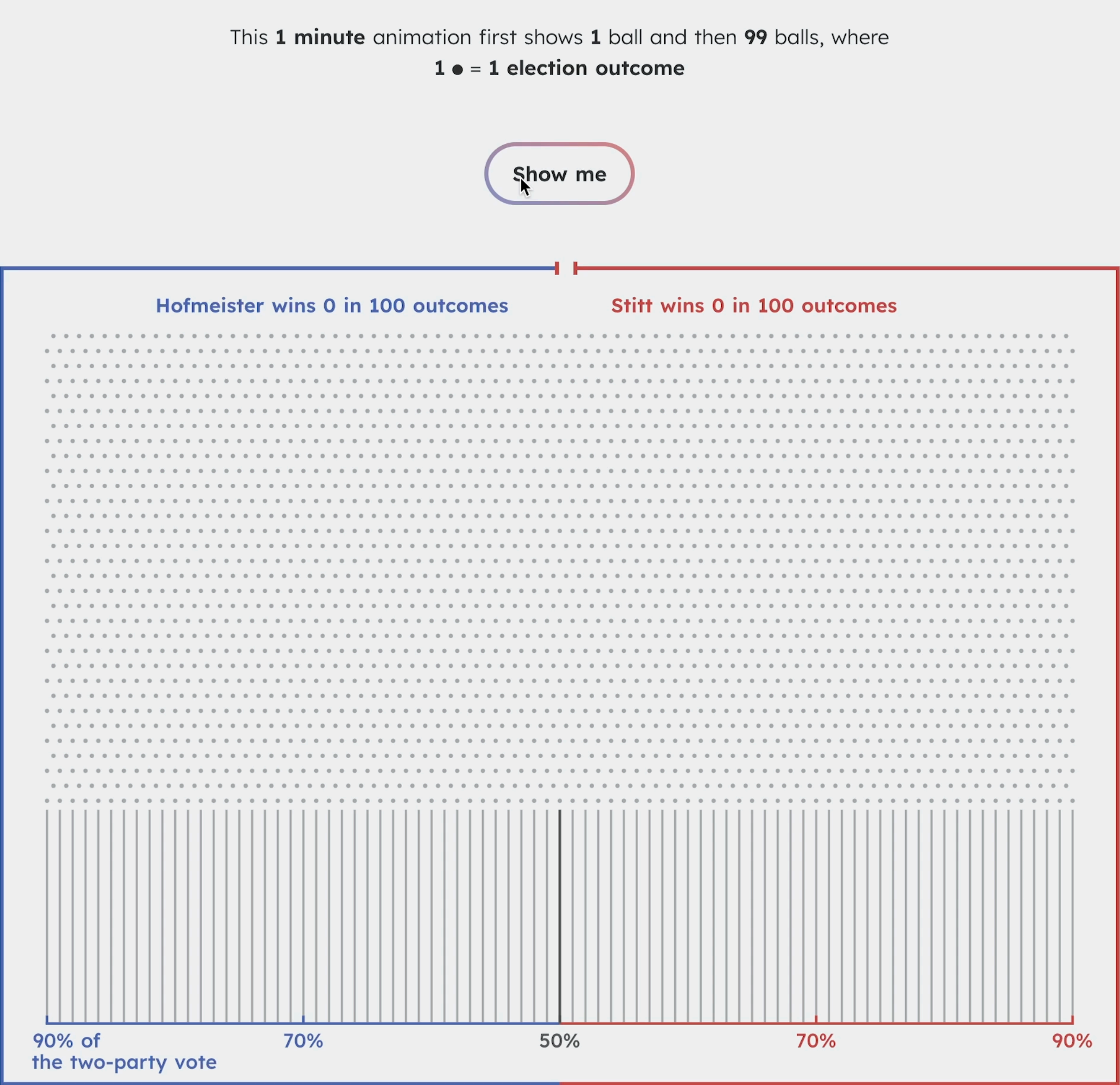
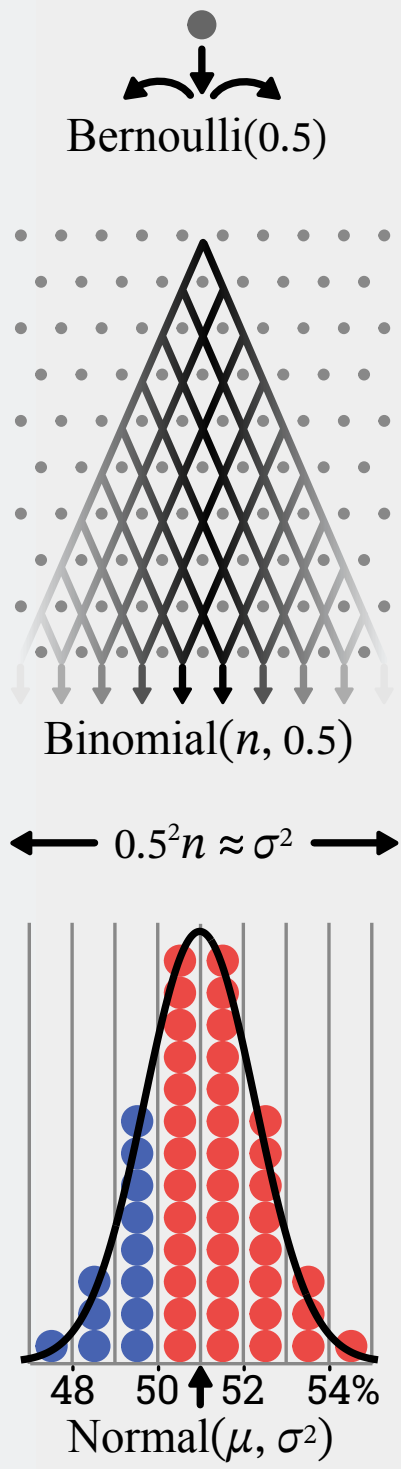
Unsure why the needle is moving....



Unscientific

We use a Plinko board to show a stats process to avoid confusion.

People: "it's unscientific."



Unscientific

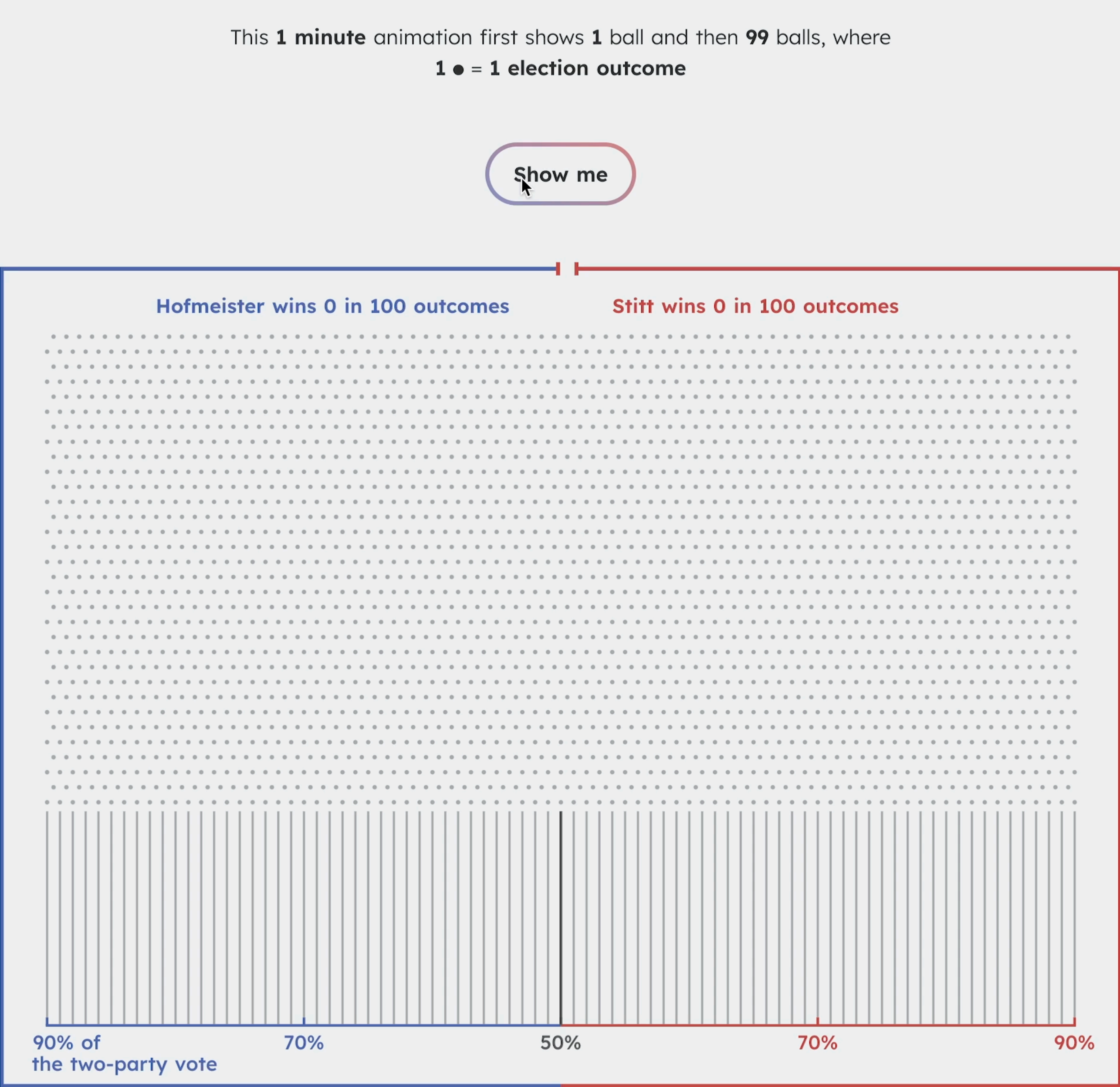
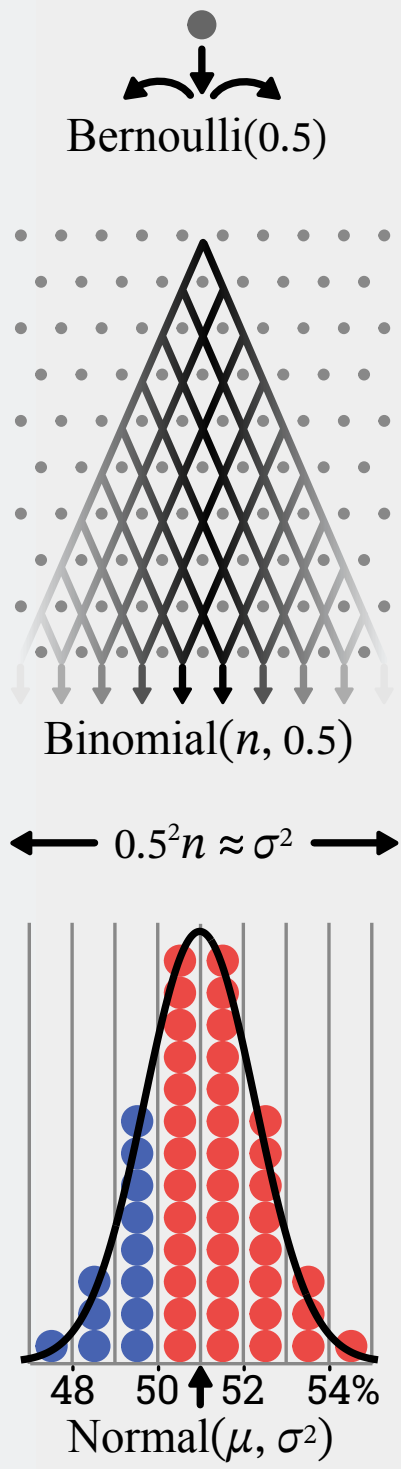


<https://www.youtube.com/watch?v=-gphdkGL4uE&t=146s>

Unscientific

We use a Plinko board to show a stats process to avoid confusion.

People: "it's unscientific."



Implementation and design

Implementation

Frame-Based Animation

Redraw the scene at regular intervals (e.g., 16ms).

Developer defines the redraw function (e.g., Processing, p5.js)

Transition-Based Animation

Specify a property value, duration, and an “easing” function.

Also called **tweening** (for “in-betweens”).

Steps computed via **interpolation**

```
step (fraction) { valnow = valstart + fraction * (valend - valstart); }
```

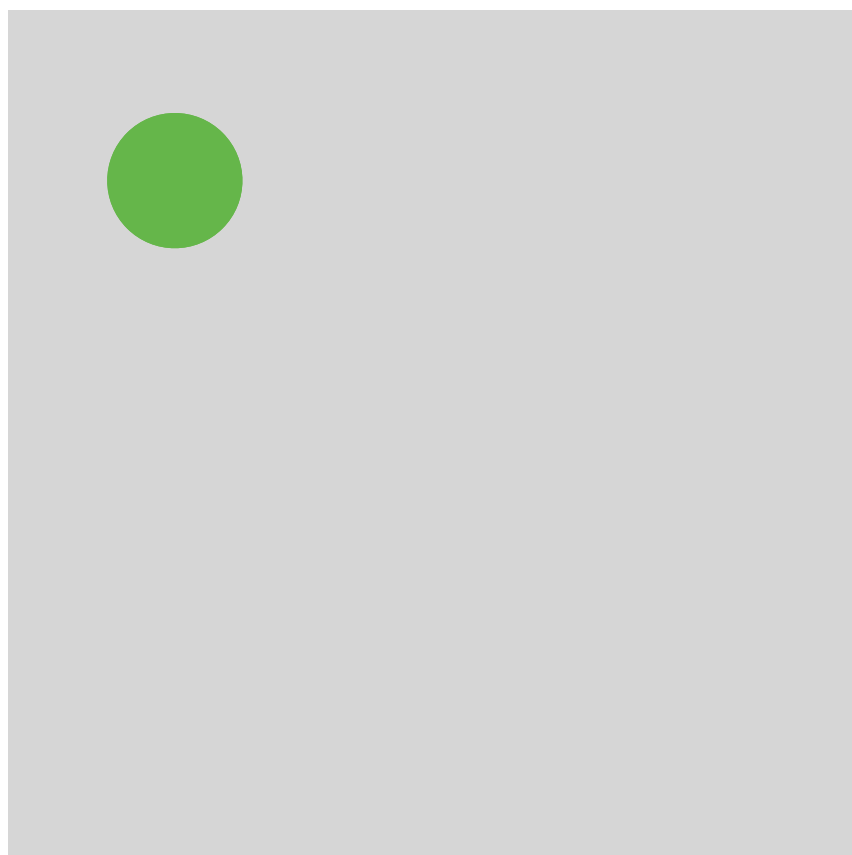
Timing & redraw managed by UI toolkit.

Frame-based animation

Redraw the scene at regular intervals (e.g., 16ms).

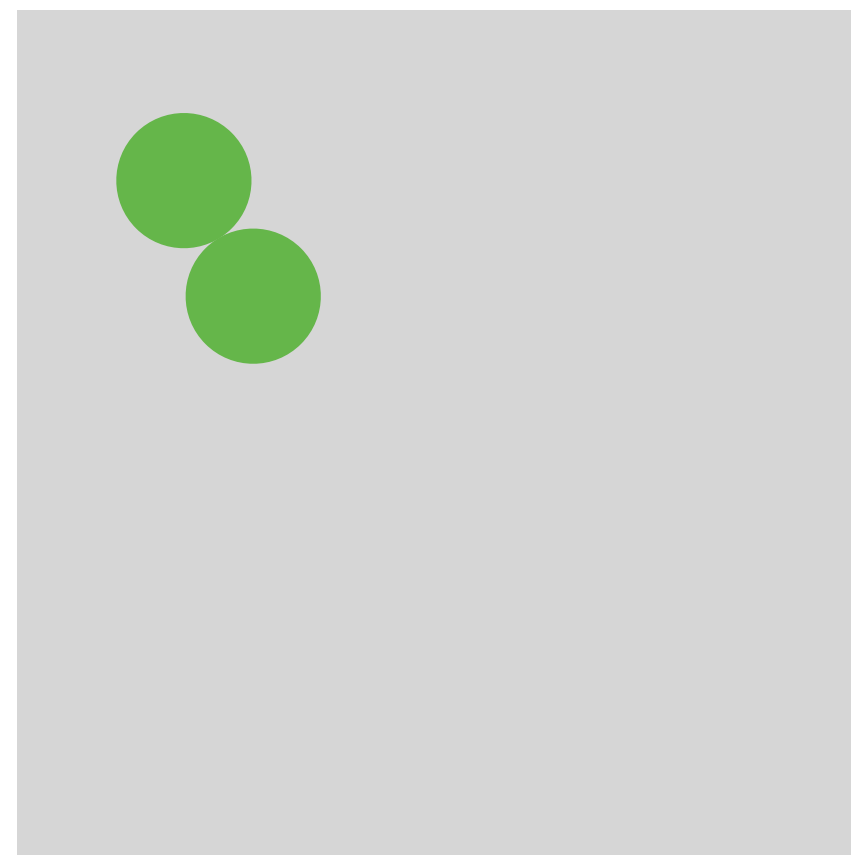
Developer defines the redraw function (e.g., Processing, p5.js).

`circle(10, 10)`



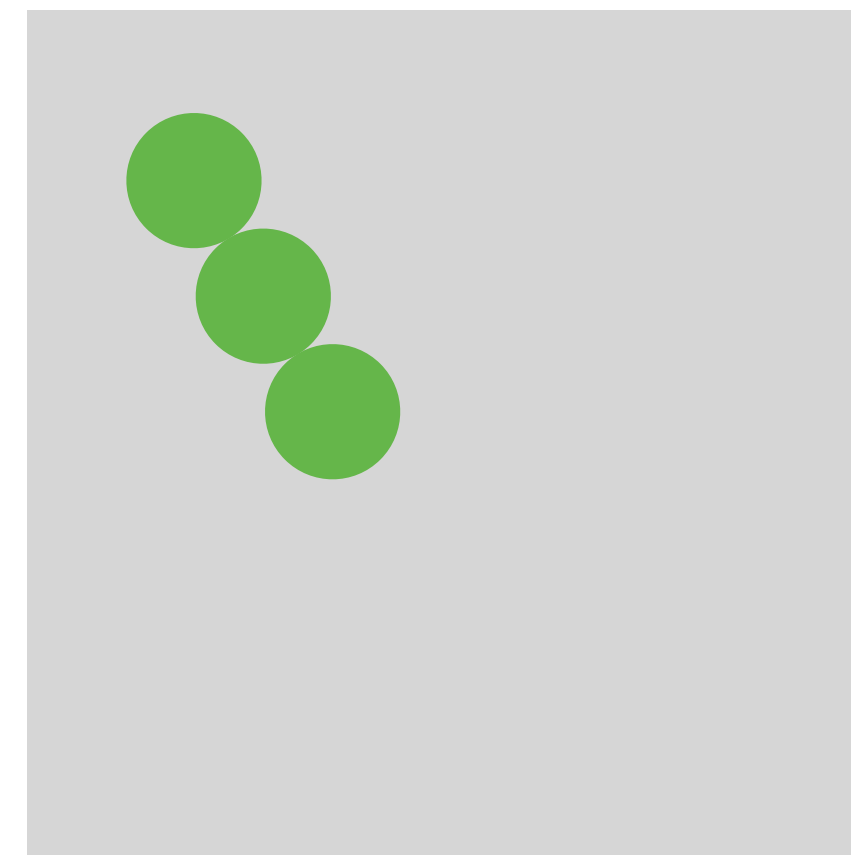
1

`circle(15, 15)`



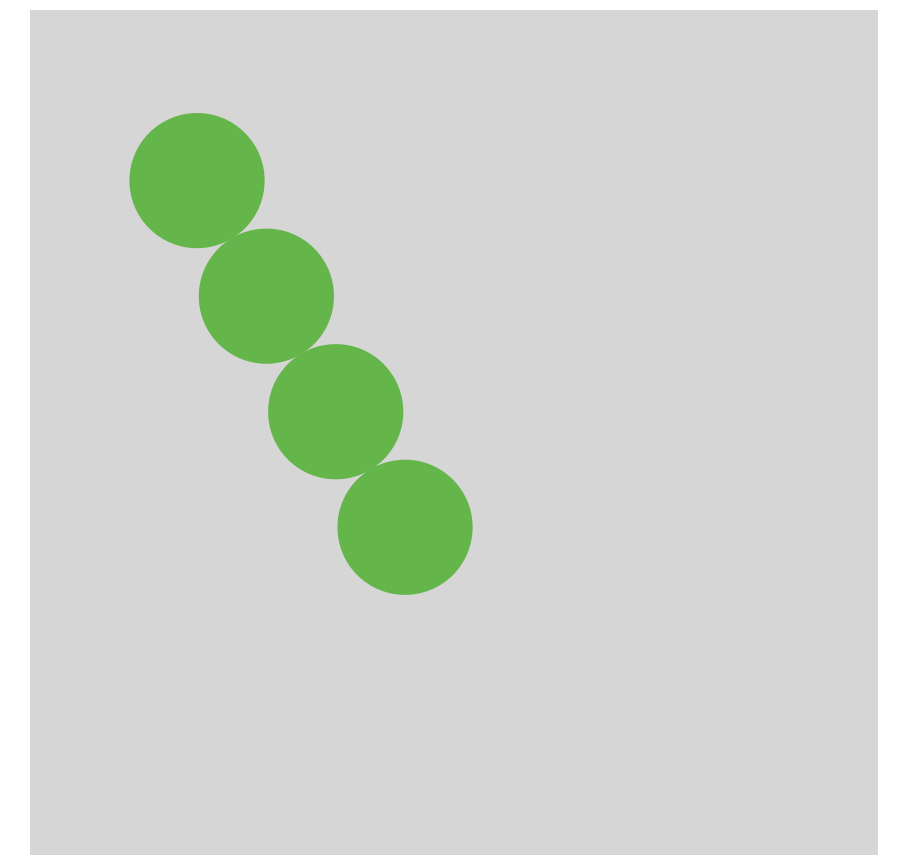
2

`circle(20, 20)`



3

`circle(25, 25)`



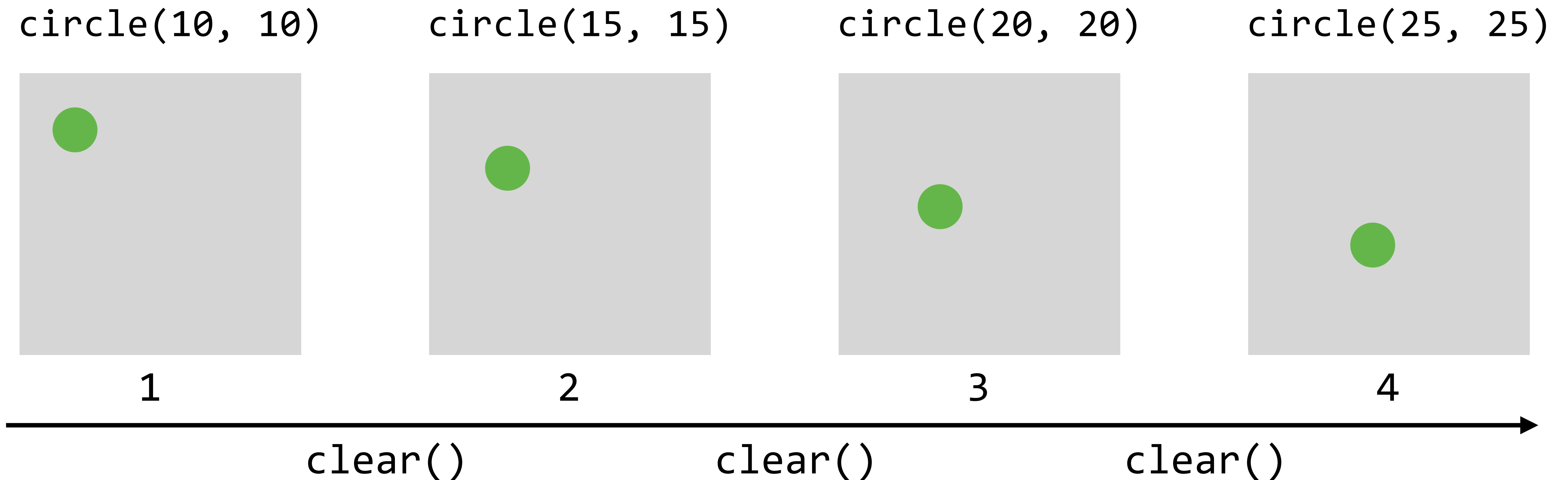
4



Frame-based animation

Redraw the scene at regular intervals (e.g., 16ms).

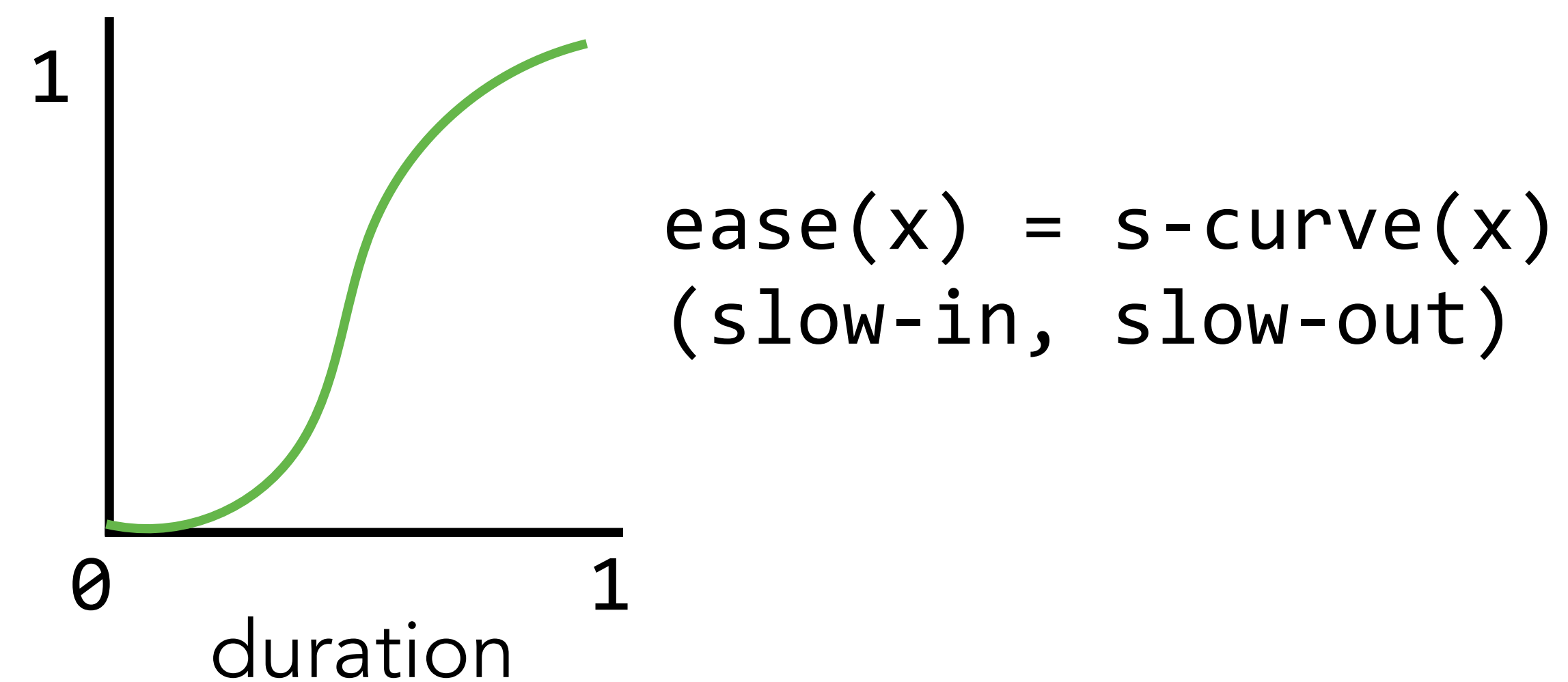
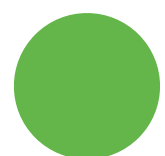
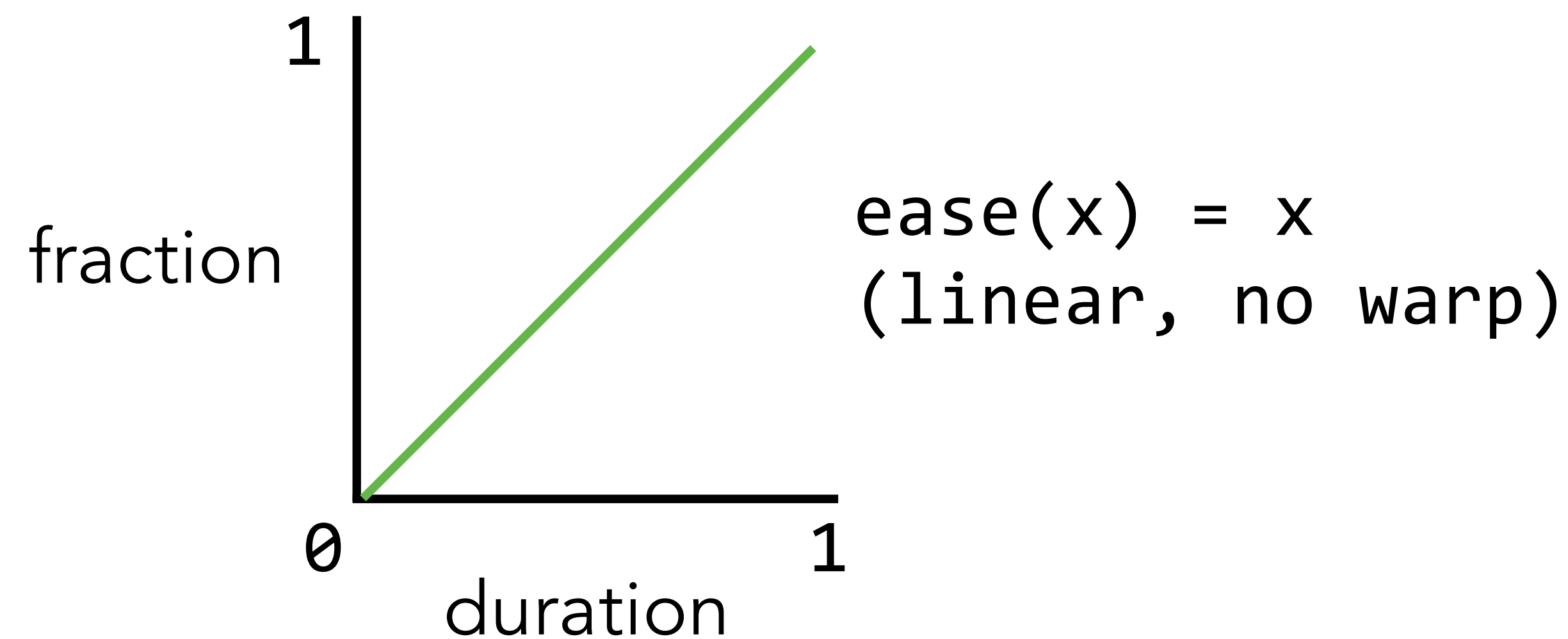
Developer defines the redraw function (e.g., Processing, p5.js).



Easing/pacing functions

Goals: Stylize animation, improve perception.

The idea is to **warp time**: as *duration* goes from start (0%) to end (100%), dynamically adjust the *interpolation fraction* using an easing function.



Easing/pacing

<https://easings.net/en>

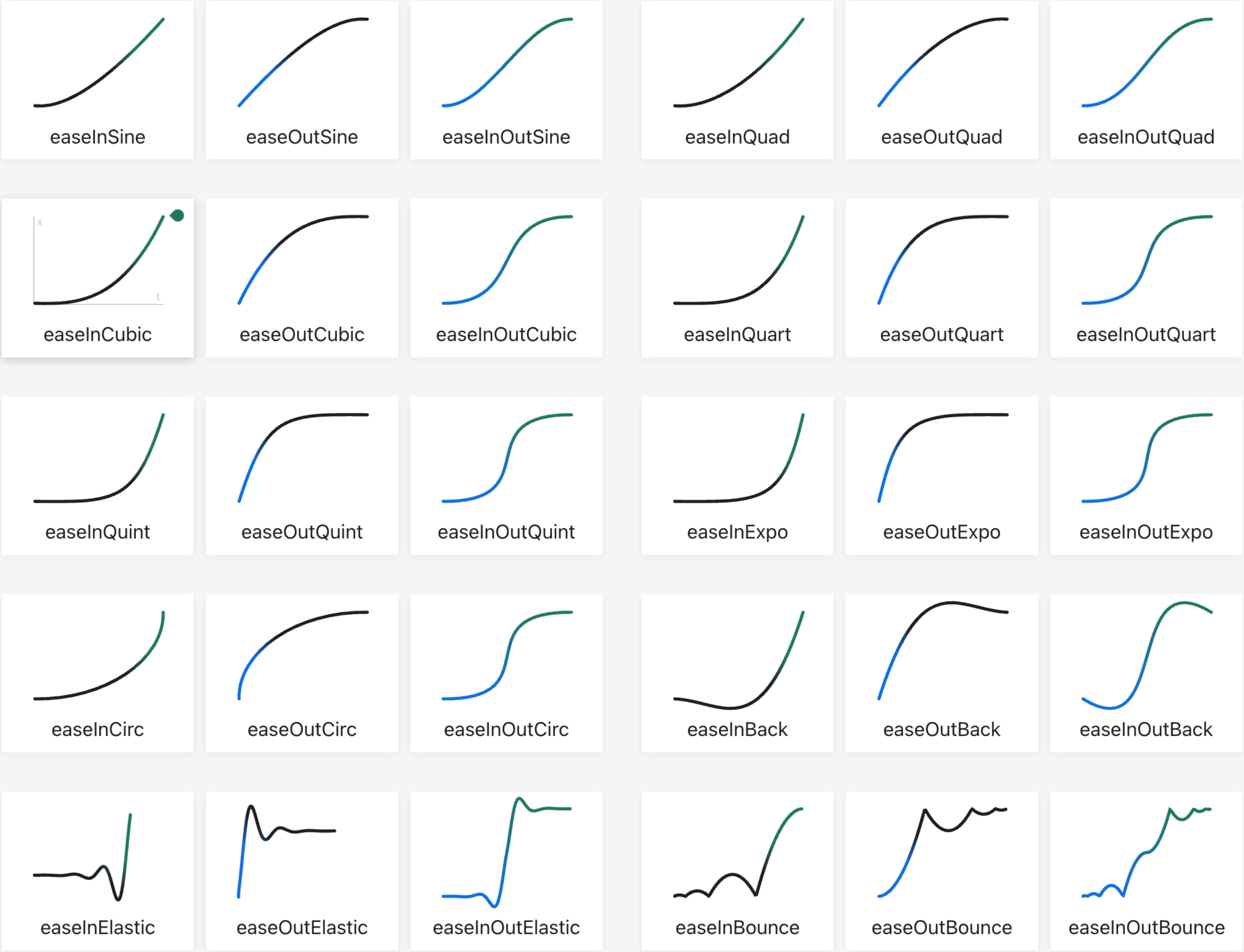
Easing functions specify the rate of change of a parameter over time.

Objects in real life don't just start and stop instantly, and almost never move at a constant speed. When we open a drawer, we first move it quickly, and slow it down as it comes out. Drop something on the floor, and it will first accelerate downwards, and then bounce back up after hitting the floor.

This page helps you choose the right easing function.

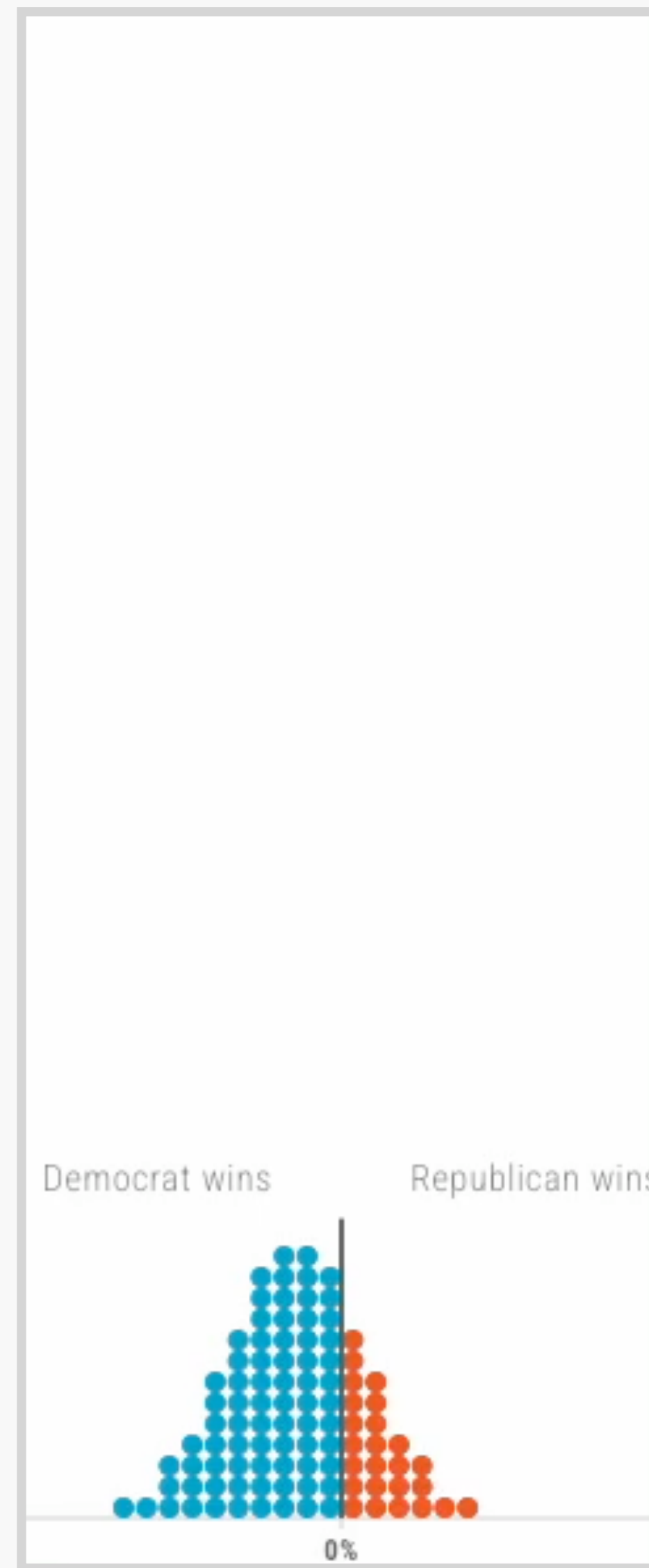
 [Open Source](#)

Help translate
site to your language

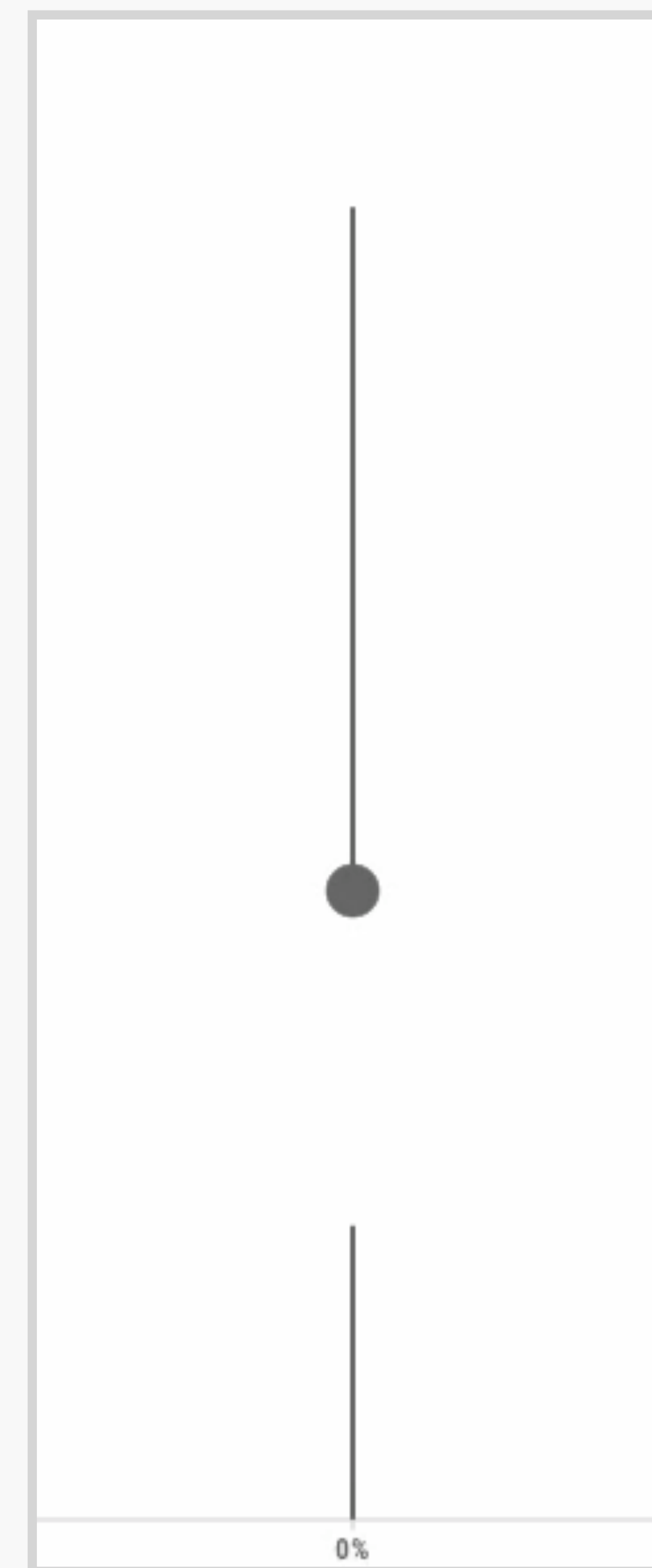


Analogy

Having a good analogy
seems to resolve some
confusion



abstract



figurative



mechanistic

Animation/Narrative

The introduction of time gives
it a narrative structure

Carefully think about what you
want to present along the timeline

Wed Lab

Labs 4-5: Interaction and Transition with D3.js

